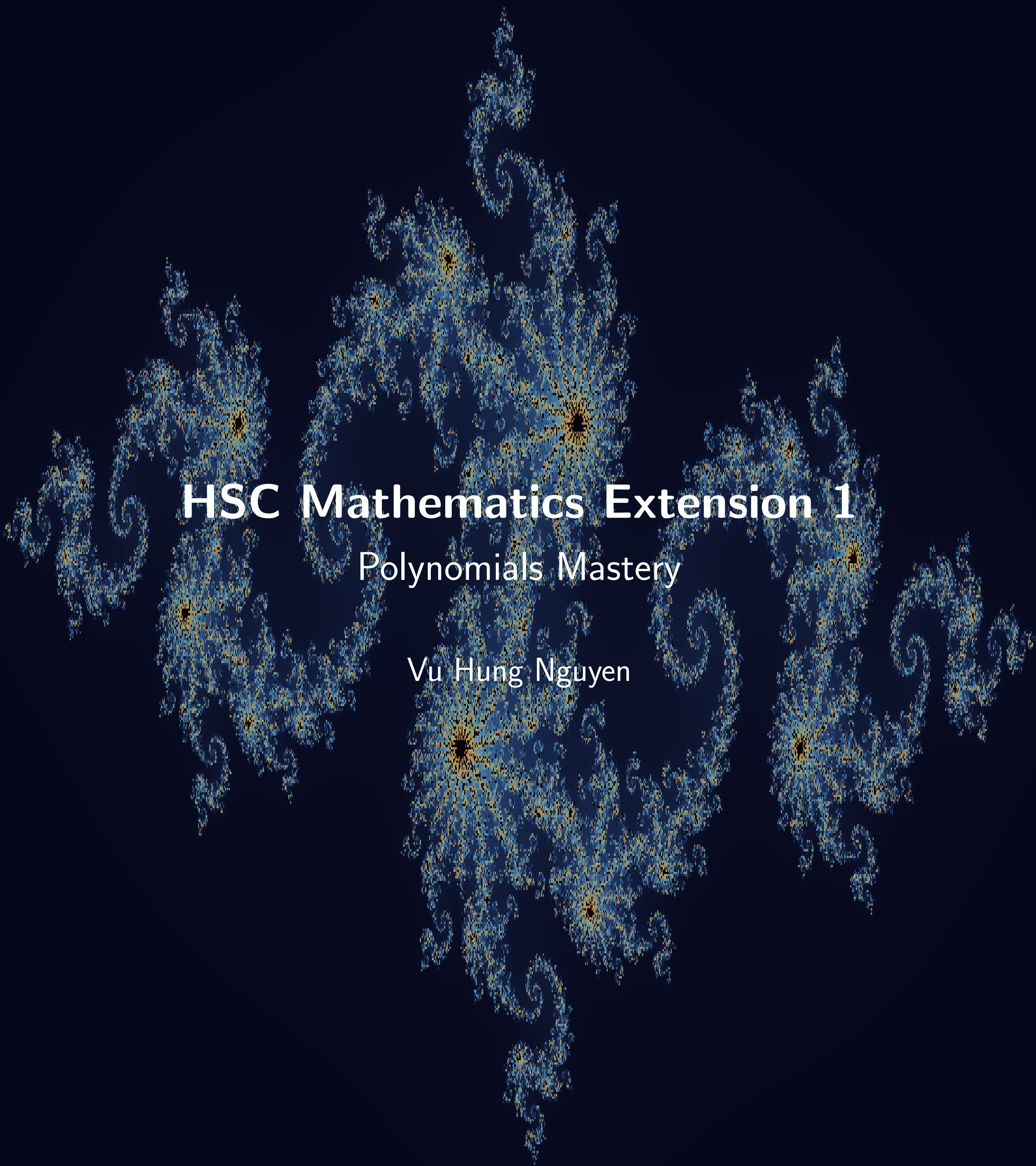

HSC Mathematics Extension 1

Polynomials Mastery

Vu Hung Nguyen

Download PDF and resources:

`https://github.com/vuhung16au/math-olympiad-ml`

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"Just because we can't find a solution it doesn't mean that there isn't one."

Andrew Wiles

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1 Introduction

1.1 Project Overview

This booklet is a growing collection of polynomial problems for NSW HSC Mathematics Extension 1 students. It covers the full polynomial strand of the Extension 1 syllabus, including the language and notation of polynomials, graphs of polynomial functions, division algorithms, the remainder and factor theorems, consequences of the factor theorem, sums and products of zeroes (roots), and multiple zeroes. The problems range from straightforward skill-drill questions to multi-step challenge problems suitable for exam preparation.

1.2 Target Audience

This resource is written for students who want focused, exam-oriented practice with polynomials in a clear, classroom-friendly format. Solutions are intentionally concise but each one highlights the key algebraic idea behind the answer. Students who work systematically through Parts 1 and 2 should gain both fluency and deeper conceptual understanding of the polynomial topics tested in Extension 1.

1.3 How to Use This Booklet

- Read the core formulas and theory summaries in this introduction first.
- Attempt each problem before reading the solution.
- Pay attention to the *type* of question: is it a factoring question, a root-extraction question, or a proof using identities?
- Look for structure: symmetry among roots, repeated factors, or relationships that allow the factor theorem to be applied efficiently.
- Use the remarks and takeaways to build pattern recognition for future HSC questions.

1.4 Extension 1 Knowledge Needed

Most problems in this booklet can be solved using standard HSC Mathematics Extension 1 knowledge. The key background material is:

- **Polynomial notation:** understanding degree, leading coefficient, constant term, and the standard form
$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0.$$
- **Equality of polynomials:** two polynomials are equal if and only if all corresponding coefficients are equal.
- **Division algorithm:** for polynomials $P(x)$ and $D(x)$ with $\deg D \le \deg P$, there exist unique $Q(x)$ and $R(x)$ with $\deg R < \deg D$ such that $P(x) = D(x)Q(x) + R(x)$.
- **Remainder theorem:** when $P(x)$ is divided by $(x-a)$, the remainder equals $P(a)$.
- **Factor theorem:** $(x-a)$ is a factor of $P(x)$ if and only if $P(a) = 0$.
- **Vieta's formulas (sums and products of roots):** for $P(x) = x^n + p_{n-1}x^{n-1} + \cdots + p_0$, the elementary symmetric functions of the roots $\alpha_1, \ldots, \alpha_n$ are expressible in terms of the coefficients.

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- **Multiple zeroes:** a is a zero of multiplicity m when $(x-a)^m \mid P(x)$ but $(x-a)^{m+1} \nmid P(x)$; equivalently, $P(a) = P'(a) = \cdots = P^{(m-1)}(a) = 0$ and $P^{(m)}(a) \neq 0$.
- **Graph behaviour near roots:** the curve crosses the x-axis at simple roots and touches (but does not cross) at even-multiplicity roots.
- **Even and odd polynomials:** P is even if $P(-x) = P(x)$ and odd if $P(-x) = -P(x)$. Even polynomials contain only even powers, while odd polynomials contain only odd powers and have constant term 0.
- **Long division and synthetic division:** systematic methods for finding quotients and remainders.
- **Sketching polynomial graphs:** leading-term behaviour, y-intercept, x-intercepts, and turning points.
- **Coefficient extraction:** in a product of sums, the coefficient of x^r can often be found by counting exponent pairs whose total is r.

1.5 Core Formulas and Ideas

Definition of a Polynomial. A **polynomial** in x of degree n is an expression of the form

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

where $a_n, a_{n-1}, \ldots, a_0$ are real (or complex) constants and $a_n \neq 0$. The highest power n is the **degree** of P, and a_n is its **leading coefficient**.

Polynomial Equality. Two polynomials are equal if and only if they have the same degree and all corresponding coefficients are equal:

$$a_n x^n + \cdots + a_0 = b_n x^n + \cdots + b_0 \iff a_k = b_k \text{ for all } k.$$

Division Algorithm for Polynomials. Given polynomials $P(x)$ and $D(x)$ with $D(x) \neq 0$ and $\deg D \leq \deg P$, there exist unique polynomials $Q(x)$ (the *quotient*) and $R(x)$ (the *remainder*) such that

$$P(x) = D(x)\,Q(x) + R(x), \qquad \deg R < \deg D \text{ (or } R = 0).$$

Remainder Theorem. When the polynomial $P(x)$ is divided by the linear factor $(x-a)$, the remainder is

$$R = P(a).$$

Factor Theorem. $(x-a)$ is a factor of $P(x)$ if and only if

$$P(a) = 0.$$

More generally, $(ax-b)$ is a factor of $P(x)$ if and only if $P\left(\frac{b}{a}\right) = 0$.

Even/Odd Test.

$$P(-x) = P(x) \iff P \text{ is even}, \qquad P(-x) = -P(x) \iff P \text{ is odd}.$$

This is especially useful when symmetry reduces the number of unknown coefficients.

1.6 Vieta's Formulas (Sums and Products of Roots)

Let $P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_0$ have roots $\alpha_1, \alpha_2, \ldots, \alpha_n$. Then:

1. **Sum of roots:**
$$\alpha_1 + \alpha_2 + \cdots + \alpha_n = -\frac{a_{n-1}}{a_n}.$$

2. **Sum of products of pairs:**
$$\sum_{i<j} \alpha_i \alpha_j = \frac{a_{n-2}}{a_n}.$$

3. **Product of all roots:**
$$\alpha_1\,\alpha_2 \cdots \alpha_n = (-1)^n\,\frac{a_0}{a_n}.$$

Quadratic case. If $P(x) = x^2 + bx + c$ has roots α, β:

$$\alpha + \beta = -b, \qquad \alpha\beta = c.$$

Cubic case. If $P(x) = x^3 + px^2 + qx + r$ has roots α, β, γ:

$$\alpha + \beta + \gamma = -p, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = q, \qquad \alpha\beta\gamma = -r.$$

1.7 Multiple Zeroes and Graph Behaviour

A number a is a zero of **multiplicity** m of the polynomial $P(x)$ if

$$(x-a)^m \mid P(x) \quad \text{but} \quad (x-a)^{m+1} \nmid P(x).$$

An equivalent characterisation using derivatives:

$$P(a) = P'(a) = \cdots = P^{(m-1)}(a) = 0, \quad P^{(m)}(a) \neq 0.$$

Graph shape at a zero:

- **Simple zero** ($m = 1$): the graph *crosses* the x-axis.
- **Double zero** ($m = 2$): the graph *touches* the x-axis and turns back (like $y = x^2$ at the origin).
- **Triple zero** ($m = 3$): the graph *flattens* and crosses the x-axis (like $y = x^3$ at the origin).

1.8 Sketching Polynomial Graphs

Key features to determine when sketching $y = P(x)$:

1. **Degree and leading coefficient:** determines end behaviour.
   - Even degree, positive leading coefficient: both ends go to $+\infty$.
   - Even degree, negative leading coefficient: both ends go to $-\infty$.
   - Odd degree, positive leading coefficient: left end $\to -\infty$, right end $\to +\infty$.
   - Odd degree, negative leading coefficient: left end $\to +\infty$, right end $\to -\infty$.
2. **y-intercept:** $P(0) = a_0$.
3. **x-intercepts:** the real roots of $P(x) = 0$, taking multiplicities into account.
4. **Turning points:** at most $n - 1$ turning points for a degree-n polynomial.

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1.9 Consequences of the Factor Theorem

- If a polynomial $P(x)$ of degree n has n distinct real roots $\alpha_1, \ldots, \alpha_n$, then

$$P(x) = a_n(x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n).$$

- A polynomial of degree n has *at most* n roots (counted with multiplicity) unless it is the zero polynomial.
- If two polynomials of degree $\leq n$ agree at $n+1$ distinct points, they must be identical.
- **Finding factors:** to factor $P(x)$, test rational candidates $\pm\frac{p}{q}$ where $p \mid a_0$ and $q \mid a_n$ (rational root theorem), then use polynomial division to find the remaining factors.

1.10 Division Techniques

Long division. Systematically divide the leading term of the dividend by the leading term of the divisor, multiply, subtract, and repeat until the degree of the remainder is less than the degree of the divisor.

Remark 1.1 (Comparing Number Division and Polynomial Division)

The same idea is used in both arithmetic and algebra:

$$\text{dividend} = \text{divisor} \times \text{quotient} + \text{remainder}.$$

Number example. Dividing 2048 by 15 gives quotient 136 and remainder 8:

$$2048 = 15 \times 136 + 8, \qquad 0 \leq 8 < 15.$$

Polynomial example. Dividing

$$x^3 + x^2 + 2x - 1 \quad \text{by} \quad x - 3$$

gives quotient $x^2 + 4x + 14$ and remainder 41:

$$x^3 + x^2 + 2x - 1 = (x - 3)(x^2 + 4x + 14) + 41.$$

Here the remainder is a constant, which is allowed because its degree is less than the degree of $x - 3$.
So the *structure* is identical:

$$\boxed{\text{Dividend} = \text{Divisor} \times \text{Quotient} + \text{Remainder.}}$$

The only difference is that for polynomials the remainder must have lower degree than the divisor.

Synthetic division (for linear divisors). A streamlined numerical method for dividing $P(x)$ by $(x - a)$; it records only the coefficients and uses the value a as the divisor.

Repeated-root test. If r is a repeated zero of $P(x)$, then

$$P(r) = 0 \qquad \text{and} \qquad P'(r) = 0.$$

This is a standard way to attack parameter questions involving double roots.

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Example. To divide $P(x) = 2x^3 - 3x^2 + x - 5$ by $(x - 2)$ using synthetic division:

$$\begin{array}{c|cccc} 2 & 2 & -3 & 1 & -5 \\ \hline & & 4 & 2 & 6 \\ \hline & 2 & 1 & 3 & 1 \end{array}$$

So $P(x) = (x - 2)(2x^2 + x + 3) + 1$, and the remainder is $P(2) = 1$. ✓

1.11 Problem-Solving Habits for This Booklet

- Always identify the *degree* and *leading coefficient* first.
- When using the factor theorem, try $x = \pm 1, \pm 2, \ldots$ before turning to the rational root theorem.
- For root-sum/product problems, write out Vieta's formulas explicitly before attempting algebra.
- For multiple-zero problems, consider taking derivatives to set up a system of equations.
- For graph sketching, mark each intercept with its multiplicity and check end behaviour.
- In proof questions, compare coefficients or use the identity principle for polynomials.

Remark 1.2

The most common polynomial mistake is forgetting to account for multiplicity: a polynomial of degree n may have *fewer* than n distinct real roots, but it always has *exactly* n roots when complex roots and multiplicities are counted together (Fundamental Theorem of Algebra).

2 Part 1: Worked Problems

2.1 Basic

Problem 2.1: Identically Equal Polynomials

Find a, b and c given that

$$a(x-2)^2 + b(x-2) + c = x^2 + x + 1 \quad \text{for all } x.$$

Hint 2.1

Expand the left-hand side and compare coefficients.

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Solution 2.1

$$a(x-2)^2 + b(x-2) + c = ax^2 + (-4a+b)x + (4a-2b+c).$$

Comparing with $x^2 + x + 1$ gives

$$a = 1, \qquad -4a + b = 1, \qquad 4a - 2b + c = 1.$$

Hence

$$\boxed{a = 1, \quad b = 5, \quad c = 7.}$$

Takeaways 2.1

- Polynomial identities are solved by coefficient comparison.
- Expanding around $(x-2)$ is still just an ordinary polynomial identity.

Problem 2.2: Odd Polynomial Coefficients

Suppose

$$P(x) = ax^5 + bx^4 + cx^3 + dx^2 + ex + f$$

is an odd function, so $P(-x) = -P(x)$ for all x. Show that $b = d = f = 0$.

Hint 2.2

Write down $P(-x)$ and compare it with $-P(x)$ term by term.

Solution 2.2

$$P(-x) = -ax^5 + bx^4 - cx^3 + dx^2 - ex + f,$$

while

$$-P(x) = -ax^5 - bx^4 - cx^3 - dx^2 - ex - f.$$

Since these are identical,

$$b = -b, \qquad d = -d, \qquad f = -f.$$

So

$$\boxed{b = d = f = 0.}$$

Takeaways 2.2

- Odd polynomials contain only odd powers of x.
- The constant term of an odd polynomial must be 0.

Problem 2.3: Central Coefficient

What is the coefficient of x^n in

$$G(x) = \left(1 + x + x^2 + \cdots + x^n\right)^2?$$

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Hint 2.3

Count the pairs (i, j) with $0 \le i, j \le n$ and $i + j = n$.

Solution 2.3

Write

$$G(x) = \left(\sum_{i=0}^{n} x^i\right)\left(\sum_{j=0}^{n} x^j\right).$$

The term x^n appears whenever $x^i \cdot x^j = x^n$, that is, whenever $i + j = n$. The possible pairs are

$$(0, n), (1, n-1), \ldots, (n, 0),$$

which gives $n + 1$ choices. Therefore the coefficient is

$$\boxed{n + 1.}$$

Takeaways 2.3

- Coefficient questions in products often reduce to counting exponent pairs.
- The coefficient of x^r is the number of ways to split r as a sum of exponents.

Problem 2.4: Easy Zeroes First

Solve

$$x^3 - x^2 - 4x + 4 = 0$$

by first factoring the left-hand side.

Hint 2.4

Factor by grouping.

Solution 2.4

$$x^3 - x^2 - 4x + 4 = x^2(x - 1) - 4(x - 1) = (x - 1)(x^2 - 4).$$

So

$$(x - 1)(x - 2)(x + 2) = 0.$$

Hence

$$\boxed{x = 1,\ 2,\ -2.}$$

Takeaways 2.4

- Before using heavier tools, try grouping or simple integer roots.
- Cubics designed for students often hide an obvious factor.

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Problem 2.5: Factors of $x^n + 1$

Is either $x+1$ or $x-1$ a factor of x^n+1, where n is a positive integer?

Hint 2.5

Use the factor theorem at $x=1$ and $x=-1$.

Solution 2.5

Let $P(x) = x^n + 1$.
For $x-1$ to be a factor we need $P(1)=0$, but

$$P(1) = 2 \neq 0.$$

So $x-1$ is never a factor.
For $x+1$ to be a factor we need $P(-1)=0$:

$$P(-1) = (-1)^n + 1.$$

If n is odd, this is 0. If n is even, this is 2.
Therefore

$$\boxed{x+1 \text{ is a factor iff } n \text{ is odd, and } x-1 \text{ is never a factor.}}$$

Takeaways 2.5

- The factor theorem turns factor questions into substitutions.
- Powers of -1 almost always split into odd and even cases.

Problem 2.6: Simple Partial Fractions

Given

$$\frac{1}{x^3-1} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1},$$

find A, B and C.

Hint 2.6

Use $x^3-1=(x-1)(x^2+x+1)$, then clear denominators.

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Solution 2.6

Since

$$x^3 - 1 = (x-1)(x^2+x+1),$$

we have

$$1 = A(x^2+x+1) + (Bx+C)(x-1).$$

Expand:

$$1 = (A+B)x^2 + (A-B+C)x + (A-C).$$

Comparing coefficients gives

$$A+B=0, \qquad A-B+C=0, \qquad A-C=1.$$

Solving,

$$\boxed{A = \frac{1}{3}, \qquad B = -\frac{1}{3}, \qquad C = -\frac{2}{3}.}$$

Takeaways 2.6

- Clear denominators before comparing coefficients.
- Partial fractions are another good place to use polynomial identities.

Problem 2.7: Partial Fractions with a Repeated Linear Factor

Let

$$P(x) = x^2(x-1)(x+1).$$

Find constants A, B, C, D such that

$$\frac{1}{P(x)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{x+1}$$

for all x for which the expression is defined.

Hint 2.7

Multiply through by $x^2(x-1)(x+1)$, then use convenient values such as $x = 0, 1, -1$ before comparing coefficients.

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Solution 2.7

We want
$$\frac{1}{x^2(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{x+1}.$$
Multiply by $x^2(x-1)(x+1)$:
$$1 = Ax(x-1)(x+1) + B(x-1)(x+1) + Cx^2(x+1) + Dx^2(x-1).$$
Now substitute convenient values:
$$x = 0 \Rightarrow 1 = -B \Rightarrow B = -1,$$
$$x = 1 \Rightarrow 1 = 2C \Rightarrow C = \frac{1}{2},$$
$$x = -1 \Rightarrow 1 = 2D \Rightarrow D = \frac{1}{2}.$$
Use $x = 2$:
$$1 = 6A + 3B + 12C + 4D.$$
Substituting $B = -1$, $C = D = \frac{1}{2}$ gives
$$1 = 6A - 3 + 6 + 2 = 6A + 5,$$
so
$$A = -\frac{2}{3}.$$
Hence
$$\boxed{A = -\frac{2}{3}, \quad B = -1, \quad C = \frac{1}{2}, \quad D = \frac{1}{2}.}$$

Takeaways 2.7

- A repeated linear factor $(x-a)^2$ contributes two terms: $\dfrac{A}{x-a} + \dfrac{B}{(x-a)^2}$.
- Substituting the roots of the denominator is usually faster than comparing every coefficient immediately.

Problem 2.8: Partial Fractions with an Irreducible Quadratic

Let
$$Q(x) = (x^2+1)(x-1)^2.$$
Find constants A, B, C, D such that
$$\frac{1}{Q(x)} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$$
for all x for which the expression is defined.

Hint 2.8

Clear denominators first. Use $x = 1$, then compare coefficients or substitute more values.

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Solution 2.8

We require

$$\frac{1}{(x^2+1)(x-1)^2} = \frac{Ax+B}{x^2+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}.$$

Multiply by $(x^2+1)(x-1)^2$:

$$1 = (Ax+B)(x-1)^2 + C(x^2+1)(x-1) + D(x^2+1).$$

Expand:

$$1 = (Ax+B)(x^2-2x+1) + C(x^3-x^2+x-1) + D(x^2+1).$$

So

$$\begin{aligned} 1 &= (A+C)x^3 + (-2A+B-C+D)x^2 \\ &\quad + (A-2B+C)x + (B-C+D). \end{aligned}$$

Compare coefficients:

$$\begin{cases} A+C=0, \\ -2A+B-C+D=0, \\ A-2B+C=0, \\ B-C+D=1. \end{cases}$$

From $A+C=0$, we get $C=-A$. Then

$$A-2B+C=0 \Rightarrow -2B=0 \Rightarrow B=0.$$

$$-2A+B-C+D=0 \Rightarrow -A+D=0 \Rightarrow D=A.$$

Finally,

$$B-C+D=1 \Rightarrow A+A=1 \Rightarrow A=\frac{1}{2}.$$

So

$$\boxed{A=\frac{1}{2}, \quad B=0, \quad C=-\frac{1}{2}, \quad D=\frac{1}{2}.}$$

$$\frac{1}{(x^2+1)(x-1)^2} = \frac{x}{2(x^2+1)} - \frac{1}{2(x-1)} + \frac{1}{2(x-1)^2}.$$

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Takeaways 2.8

- Integrals of the form $\int \frac{1}{(x^2+1)(x-1)^2}\,dx$ can be evaluated using partial fractions.
- An irreducible quadratic factor (ax^2+bx+c) contributes a numerator of the form $Mx+N$ in the fully general case.
- General rule: if
$$P(x) = \prod (L_i(x))^{r_i} \prod (Q_j(x))^{s_j},$$
where each L_i is linear and each Q_j is an irreducible quadratic, then
$$\frac{1}{P(x)}$$
is expanded using
$$\frac{A_1}{L_i} + \frac{A_2}{L_i^2} + \cdots + \frac{A_{r_i}}{L_i^{r_i}}$$
for each repeated linear factor, and
$$\frac{M_1x+N_1}{Q_j} + \frac{M_2x+N_2}{Q_j^2} + \cdots + \frac{M_{s_j}x+N_{s_j}}{Q_j^{s_j}}$$
for each repeated irreducible quadratic factor.

2.2 Medium

Problem 2.9: Parity and Derivatives

Suppose $P(x)$ is a polynomial. Prove or disprove:

$P(x)$ is even if and only if $P'(x)$ is odd.

Hint 2.9

Prove each direction separately. For the reverse direction, examine $P(x) - P(-x)$.

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Solution 2.9

If P is even, then $P(-x) = P(x)$. Differentiating gives

$$-P'(-x) = P'(x),$$

so

$$P'(-x) = -P'(x).$$

Hence P' is odd.
Conversely, suppose P' is odd, so $P'(-x) = -P'(x)$. Let

$$F(x) = P(x) - P(-x).$$

Then

$$F'(x) = P'(x) + P'(-x) = 0.$$

So F is constant. But $F(0) = 0$, hence $F(x) = 0$ for all x. Therefore

$$P(x) = P(-x),$$

so P is even.
The statement is **true**.

Takeaways 2.9

- Differentiating a symmetry identity is often the quickest proof.
- If a polynomial has derivative 0, it must be constant.

Problem 2.10: Behaviour Near Repeated Zeroes

Let

$$P(x) = (x+1)^3 x^2 (x-1)^2.$$

Discuss the behaviour of $P(x)$ around $-\infty$, $+\infty$, -1, 0 and 1. Summarise it in a table and hence sketch the graph roughly.

Hint 2.10

Use the degree and leading coefficient for end behaviour, and multiplicity for the local shape at each zero.

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Solution 2.10

The degree is 7 and the leading coefficient is positive, so

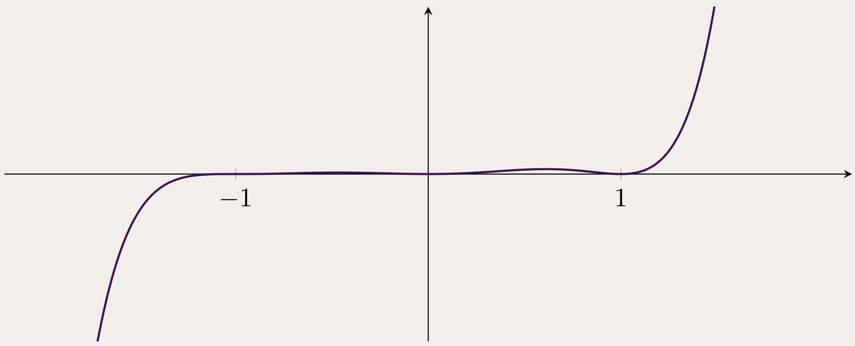
$$x \to -\infty \Rightarrow P(x) \to -\infty, \qquad x \to +\infty \Rightarrow P(x) \to +\infty.$$

The real zeroes are

$$x = -1 \text{ (multiplicity 3)}, \quad x = 0 \text{ (multiplicity 2)}, \quad x = 1 \text{ (multiplicity 2)}.$$

Hence

Point	Behaviour
$x \to -\infty$	$P(x) \to -\infty$
$x = -1$	crosses with flattening
$x = 0$	touches and turns
$x = 1$	touches and turns
$x \to +\infty$	$P(x) \to +\infty$



Takeaways 2.10

- Odd multiplicity means the graph crosses the axis.
- Even multiplicity means the graph touches and turns.

Problem 2.11: Same Axes, Different Quadratics

Sketch the following graphs on the same xy-diagram, using a different colour for each:

$$P(x) = x^2, \qquad Q(x) = (x+1)(x-2), \qquad R(x) = (x-2)^2, \qquad S(x) = 4 - x^2.$$

Hint 2.11

Find the vertex and intercepts of each quadratic first.

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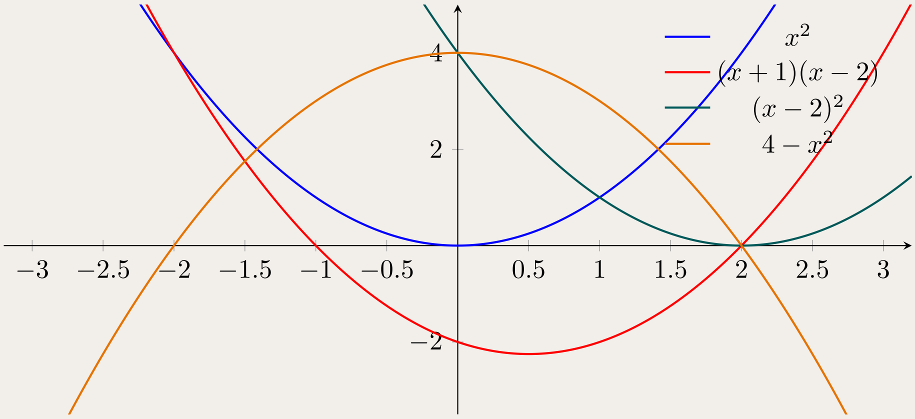
Solution 2.11

Useful features:

$$P(x) = x^2 \text{ has vertex } (0,0),$$
$$Q(x) = x^2 - x - 2 \text{ has roots } -1, 2 \text{ and vertex } \left(\frac{1}{2}, -\frac{9}{4}\right),$$
$$R(x) = (x-2)^2 \text{ has vertex } (2,0),$$
$$S(x) = 4 - x^2 \text{ has vertex } (0,4) \text{ and roots } \pm 2.$$



Takeaways 2.11

- Related quadratics are easiest to compare on the same set of axes.
- Vertex form and factor form both give quick sketching information.

Problem 2.12: An Odd Cubic from Two Conditions

$P(x)$ is an odd polynomial of degree 3. It has $x+2$ as a factor, and when divided by $x-3$ the remainder is 2. Find $P(x)$.

Hint 2.12

An odd cubic has the form $ax^3 + bx$.

Solution 2.12

Let
$$P(x) = ax^3 + bx.$$
Since $x + 2$ is a factor,
$$P(-2) = 0 \Rightarrow -8a - 2b = 0 \Rightarrow 4a + b = 0.$$
Since the remainder on division by $x - 3$ is 2,
$$P(3) = 2 \Rightarrow 27a + 3b = 2.$$
Using $b = -4a$,
$$27a - 12a = 2 \Rightarrow 15a = 2 \Rightarrow a = \frac{2}{15}.$$
Then
$$b = -\frac{8}{15}.$$
So
$$\boxed{P(x) = \frac{2}{15}x^3 - \frac{8}{15}x = \frac{2}{15}x(x + 2)(x - 2).}$$

Takeaways 2.12

- Symmetry conditions reduce the number of unknown coefficients.
- Factor and remainder conditions become linear equations.

Problem 2.13: Quadratic Roots and Symmetric Expressions

If α and β are the roots of
$$x^2 - 5x + 1 = 0,$$
find:

(i) $\left(\alpha + \frac{2}{\alpha}\right)\left(\beta + \frac{2}{\beta}\right)$

(ii) $\alpha^2 + \beta^2$

(iii) $\alpha^3 + \beta^3$

Hint 2.13

Use $\alpha + \beta = 5$ and $\alpha\beta = 1$.

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Solution 2.13

By Vieta,

$$\alpha + \beta = 5, \qquad \alpha\beta = 1.$$

(ii)

$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = 25 - 2 = 23.$$

(iii)

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 125 - 15 = 110.$$

(i)

$$\begin{aligned}\left(\alpha + \frac{2}{\alpha}\right)\left(\beta + \frac{2}{\beta}\right) &= \alpha\beta + 2\frac{\alpha}{\beta} + 2\frac{\beta}{\alpha} + \frac{4}{\alpha\beta} \\ &= 1 + 2(\alpha^2 + \beta^2) + 4 \\ &= 1 + 46 + 4 = 51.\end{aligned}$$

Hence

$$\boxed{51, \quad 23, \quad 110.}$$

Takeaways 2.13

- Vieta is the fastest route for symmetric expressions in the roots.
- Compute the simpler expressions first and reuse them.

Problem 2.14: A Cubic Root Product Sum

Let α, β, γ be the roots of

$$x^3 - 5x^2 + 5x - 1 = 0.$$

Find

(i) $\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2$

(ii) $\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3$

Hint 2.14

Factor out $\alpha\beta\gamma$. Then use $\alpha + \beta + \gamma$ and $\alpha^2 + \beta^2 + \gamma^2$.

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Solution 2.14

(i)

$$\alpha^2\beta\gamma + \alpha\beta^2\gamma + \alpha\beta\gamma^2 = \alpha\beta\gamma(\alpha + \beta + \gamma).$$

By Vieta,

$$\alpha + \beta + \gamma = 5, \qquad \alpha\beta\gamma = 1.$$

So the value is

$$\boxed{5.}$$

(ii)

$$\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3 = \alpha\beta\gamma(\alpha^2 + \beta^2 + \gamma^2).$$

Now

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha).$$

By Vieta for

$$\alpha + \beta + \gamma = 5, \qquad \alpha\beta + \beta\gamma + \gamma\alpha = 5, \qquad \alpha\beta\gamma = 1.$$

So

$$\alpha^2 + \beta^2 + \gamma^2 = 25 - 10 = 15.$$

Hence

$$\alpha^3\beta\gamma + \alpha\beta^3\gamma + \alpha\beta\gamma^3 = 1 \cdot 15 = \boxed{15.}$$

Takeaways 2.14

- Factor the target expression before applying Vieta.
- Many cubic root-sum questions become one line after the right factorisation.
- Symmetric root expressions are usually the ones that give a single definite answer.

Problem 2.15: The Opposite Roots

Consider the cubic polynomial

$$P(x) = x^3 - 4x^2 + kx + 16,$$

where k is a real constant. It is known that the sum of two of the roots of $P(x) = 0$ is zero.

(i) Find the value of k.

(ii) Hence find all three roots of the polynomial.

Hint 2.15

If two roots sum to zero, write them as α and $-\alpha$. Then use the sum of roots first.

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Solution 2.15

Let the roots be α, β, γ, with

$$\alpha + \beta = 0.$$

So we may write the roots as

$$\alpha,\ -\alpha,\ \gamma.$$

(i) By Vieta, the sum of the roots is

$$\alpha + (-\alpha) + \gamma = 4,$$

so

$$\gamma = 4.$$

Hence $x = 4$ is a root, and therefore

$$P(4) = 0.$$

Substitute into the polynomial:

$$4^3 - 4 \cdot 4^2 + 4k + 16 = 0.$$

This gives

$$64 - 64 + 4k + 16 = 0 \Rightarrow 4k = -16 \Rightarrow \boxed{k = -4.}$$

(ii) Now

$$P(x) = x^3 - 4x^2 - 4x + 16.$$

Factor by grouping:

$$P(x) = x^2(x-4) - 4(x-4) = (x^2 - 4)(x - 4).$$

So

$$P(x) = (x-2)(x+2)(x-4).$$

Hence the three roots are

$$\boxed{4,\ 2,\ -2.}$$

Takeaways 2.15

- The phrase "two roots sum to zero" should immediately suggest a pair of opposite roots.
- Once one root is identified, the factor theorem is often the fastest way to find the parameter.
- Before using division, always check whether the cubic factors neatly by grouping.

Problem 2.16: The Shifting Exponents

Consider the polynomial

$$P(x) = x^n + 2x^{n-1} + k,$$

where n is an integer greater than 1, and k is a real constant.

(i) If n is odd, find the value of k such that $(x + 1)$ is a factor of $P(x)$.

(ii) Suppose instead that n is even and $k = -3$. Find the remainder when $P(x)$ is divided by $x^2 - 1$.

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Hint 2.16

Use $x = -1$ for part (i). For part (ii), write the remainder as $ax + b$ and use the roots $x = 1$ and $x = -1$ of $x^2 - 1$.

Solution 2.16

(i) If $(x + 1)$ is a factor, then by the factor theorem

$$P(-1) = 0.$$

So

$$(-1)^n + 2(-1)^{n-1} + k = 0.$$

Since n is odd,

$$(-1)^n = -1, \qquad (-1)^{n-1} = 1.$$

Hence

$$-1 + 2 + k = 0 \Rightarrow 1 + k = 0,$$

so

$$\boxed{k = -1.}$$

(ii) Now let n be even and $k = -3$, so

$$P(x) = x^n + 2x^{n-1} - 3.$$

When dividing by $x^2 - 1$, the remainder has the form

$$R(x) = ax + b.$$

Since $x^2 - 1 = (x - 1)(x + 1)$, we use $x = 1$ and $x = -1$.
At $x = 1$:

$$P(1) = 1 + 2 - 3 = 0,$$

so

$$R(1) = a + b = 0.$$

At $x = -1$: since n is even, $(-1)^n = 1$ and $(-1)^{n-1} = -1$, so

$$P(-1) = 1 + 2(-1) - 3 = -4.$$

Thus

$$R(-1) = -a + b = -4.$$

Solving

$$a + b = 0, \qquad -a + b = -4$$

gives

$$a = 2, \qquad b = -2.$$

Therefore the remainder is

$$\boxed{2x - 2.}$$

Takeaways 2.16

- When substituting $x = -1$, the parity of the exponent completely controls the sign.
- For division by a quadratic, use a general linear remainder.
- If the divisor factors nicely, substituting its roots is usually the fastest way to determine the remainder.

Problem 2.17: The Parity of Powers

Consider the polynomial

$$P(x) = x^n + c^n,$$

where n is a positive integer and c is a non-zero real constant.

(i) By considering the cases where n is odd and even separately, determine whether $(x + c)$ and $(x - c)$ are factors of $P(x)$.

(ii) Suppose n is odd. Find an expression for the remainder when $P(x)$ is divided by $x^2 - c^2$, in terms of c and n.

Hint 2.17

Test $(x - c)$ using $P(c)$ and test $(x + c)$ using $P(-c)$. For part (ii), let the remainder be $Ax + B$.

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Solution 2.17

(i) To test $(x-c)$, evaluate

$$P(c) = c^n + c^n = 2c^n.$$

Since $c \neq 0$, this is not zero. So

$$\boxed{(x-c) \text{ is never a factor.}}$$

To test $(x+c)$, evaluate

$$P(-c) = (-c)^n + c^n.$$

If n is even, then $(-c)^n = c^n$, so

$$P(-c) = 2c^n \neq 0.$$

If n is odd, then $(-c)^n = -c^n$, so

$$P(-c) = 0.$$

Therefore

$$\boxed{(x+c) \text{ is a factor if and only if } n \text{ is odd.}}$$

(ii) Now suppose n is odd. When dividing by $x^2 - c^2$, the remainder has the form

$$R(x) = Ax + B.$$

Since

$$x^2 - c^2 = (x-c)(x+c),$$

we use $x = c$ and $x = -c$.
At $x = -c$, part (i) gives

$$P(-c) = 0,$$

so

$$R(-c) = -Ac + B = 0.$$

Hence

$$B = Ac.$$

At $x = c$,

$$P(c) = 2c^n,$$

so

$$R(c) = Ac + B = 2c^n.$$

Substitute $B = Ac$:

$$2Ac = 2c^n.$$

Since $c \neq 0$,

$$A = c^{n-1}.$$

Then

$$B = Ac = c^n.$$

So the remainder is

$$\boxed{R(x) = c^{n-1}x + c^n = c^{n-1}(x+c).}$$

Takeaways 2.17

- Powers of $-c$ must always be split into even and odd cases.
- Once the divisor factors, its roots give the fastest route to the linear remainder.
- Part (ii) becomes easier because the result from part (i) gives one of the remainder equations immediately.

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Problem 2.18: The Cascading Powers

The cubic equation

$$x^3 - 3x^2 + 4x - 5 = 0$$

has roots α, β, γ.

(i) Find the exact value of $\alpha^2 + \beta^2 + \gamma^2$.

(ii) By noting that $P(\alpha) = 0$, or otherwise, find the value of $\alpha^3 + \beta^3 + \gamma^3$.

(iii) Hence evaluate $\alpha^4 + \beta^4 + \gamma^4$.

(iv) Let

$$S_k = \alpha^k + \beta^k + \gamma^k.$$

Express S_k in terms of S_{k-1}, S_{k-2} and S_{k-3}.

Hint 2.18

Start with Vieta. For higher powers, rewrite the equation as a recurrence for powers of a root.

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Solution 2.18

By Vieta,

$$\alpha+\beta+\gamma=3, \qquad \alpha\beta+\beta\gamma+\gamma\alpha=4, \qquad \alpha\beta\gamma=5.$$

(i) Use

$$\alpha^2+\beta^2+\gamma^2=(\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha).$$

So

$$\alpha^2+\beta^2+\gamma^2=3^2-2(4)=9-8=\boxed{1.}$$

(ii) Since α is a root,

$$\alpha^3-3\alpha^2+4\alpha-5=0,$$

so

$$\alpha^3=3\alpha^2-4\alpha+5.$$

Similarly,

$$\beta^3=3\beta^2-4\beta+5, \qquad \gamma^3=3\gamma^2-4\gamma+5.$$

Adding,

$$\alpha^3+\beta^3+\gamma^3=3(\alpha^2+\beta^2+\gamma^2)-4(\alpha+\beta+\gamma)+15.$$

Hence

$$\alpha^3+\beta^3+\gamma^3=3(1)-4(3)+15=\boxed{6.}$$

(iii) Multiply the original equation by x:

$$x^4-3x^3+4x^2-5x=0.$$

So for each root,

$$\alpha^4=3\alpha^3-4\alpha^2+5\alpha,$$

and similarly for β,γ. Summing,

$$\alpha^4+\beta^4+\gamma^4=3(\alpha^3+\beta^3+\gamma^3)-4(\alpha^2+\beta^2+\gamma^2)+5(\alpha+\beta+\gamma).$$

Therefore

$$\alpha^4+\beta^4+\gamma^4=3(6)-4(1)+5(3)=18-4+15=\boxed{29.}$$

(iv) Since each root satisfies

$$x^3-3x^2+4x-5=0,$$

we have, for any root $r\in\{\alpha,\beta,\gamma\}$,

$$r^3=3r^2-4r+5.$$

Multiply by r^{k-3}:

$$r^k=3r^{k-1}-4r^{k-2}+5r^{k-3}.$$

Now do this for $r=\alpha,\beta,\gamma$ and add. Then

$$S_k=3S_{k-1}-4S_{k-2}+5S_{k-3}.$$

So the required recurrence is

$$\boxed{S_k=3S_{k-1}-4S_{k-2}+5S_{k-3} \qquad (k\geq 3).}$$

Takeaways 2.18

- Vieta gives the starting symmetric sums.
- A polynomial equation satisfied by each root can be reused to generate higher powers.
- When summing three root equations, remember to add the constant term three times.
- The same idea gives a recurrence for all higher power sums S_k.

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Problem 2.19: Sketching a Cubic with Derivatives

Sketch the graph of

$$f(x) = x^3 - 2x^2 - 3x.$$

Start by:

- finding $f'(x)$ and solving $f'(x) = 0$,
- analysing the sign of $f'(x)$ and the turning points,
- summarising the behaviour in a table,
- then drawing the graph.

Hint 2.19

Factor both $f(x)$ and $f'(x)$. The derivative tells you where the graph is increasing or decreasing.

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Solution 2.19

Differentiate:

$$f'(x) = 3x^2 - 4x - 3.$$

Solve

$$3x^2 - 4x - 3 = 0.$$

Using the quadratic formula,

$$x = \frac{4 \pm \sqrt{16+36}}{6} = \frac{4 \pm 2\sqrt{13}}{6} = \frac{2 \pm \sqrt{13}}{3}.$$

Since $f'(x)$ is an upward-opening quadratic,

$$f'(x) > 0 \text{ for } x < \frac{2-\sqrt{13}}{3} \quad \text{and} \quad x > \frac{2+\sqrt{13}}{3},$$

and

$$f'(x) < 0 \text{ for } \frac{2-\sqrt{13}}{3} < x < \frac{2+\sqrt{13}}{3}.$$

Therefore:

- f is increasing, then decreasing, then increasing;
- there is a local maximum at $x = \dfrac{2-\sqrt{13}}{3}$;
- there is a local minimum at $x = \dfrac{2+\sqrt{13}}{3}$.

Also

$$f(x) = x(x-3)(x+1),$$

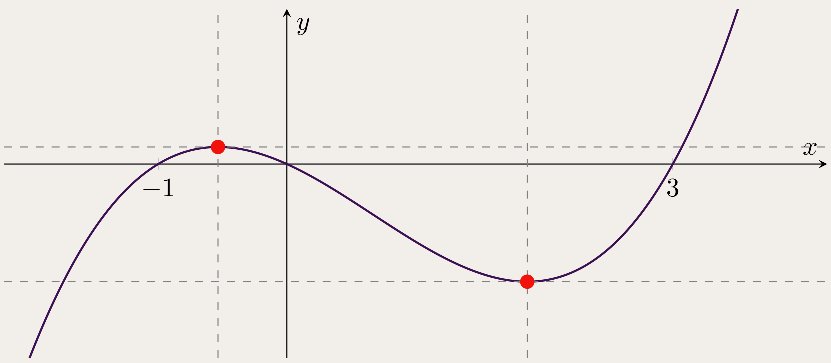
so the x-intercepts are

$$x = -1,\ 0,\ 3.$$

The behaviour table is:

x	$-\infty$	$\frac{2-\sqrt{13}}{3}$		$\frac{2+\sqrt{13}}{3}$	$+\infty$
$f'(x)$	$+$	0	$-$	0	$+$
$f(x)$	$\nearrow$	local max	$\searrow$	local min	$\nearrow$

A rough sketch is:



Takeaways 2.19

- To sketch a cubic accurately, use both the intercepts and the derivative.
- The sign of $f'(x)$ tells you where the graph is increasing or decreasing.
- A local maximum occurs where $f'(x)$ changes from positive to negative; a local minimum occurs where it changes from negative to positive.

Problem 2.20: Divisibility by $(x+1)^2$

Consider the polynomial

$$P(x) = ax^8 + bx^7 + 1,$$

where a and b are non-zero real numbers. Find the values of a and b for which $P(x)$ is divisible by $(x+1)^2$.

Hint 2.20

If $(x+1)^2$ divides $P(x)$, then $x = -1$ is a repeated root. So use $P(-1) = 0$ and $P'(-1) = 0$.

Solution 2.20

For $(x+1)^2$ to divide $P(x)$, we need

$$P(-1) = 0 \qquad \text{and} \qquad P'(-1) = 0.$$

Now

$$P(-1) = a - b + 1,$$

so

$$a - b + 1 = 0. \tag{1}$$

Also

$$P'(x) = 8ax^7 + 7bx^6,$$

hence

$$P'(-1) = -8a + 7b = 0. \tag{2}$$

From (2),

$$7b = 8a \Rightarrow b = \frac{8a}{7}.$$

Substitute into (1):

$$a - \frac{8a}{7} + 1 = 0 \Rightarrow -\frac{a}{7} + 1 = 0 \Rightarrow a = 7.$$

Then

$$b = \frac{8a}{7} = 8.$$

Therefore

$$\boxed{a = 7, \qquad b = 8.}$$

Takeaways 2.20

- If $(x-r)^2$ is a factor, then both $P(r)$ and $P'(r)$ must be zero.
- Repeated-factor questions usually become a pair of simultaneous equations in the unknown coefficients.

Problem 2.21: A Functional Equation for a Polynomial

Let $P(x)$ be a polynomial with real coefficients such that for all real x,

$$(x-3)P(x+1) = (x-1)P(x).$$

You are given that $P(0) = 16$.

(i) By choosing appropriate values of x, find two roots of $P(x)$.

(ii) Hence, express $P(x)$ in terms of another polynomial $Q(x)$, and prove that $Q(x)$ must be a constant.

(iii) Determine the explicit polynomial $P(x)$.

Hint 2.21

- **Part (i):** Substitute values for x that zero out the $(x-3)$ and $(x-1)$ multipliers.
- **Part (ii):** Apply the Factor Theorem using your roots, substitute the general form back into the equation, and simplify to show $Q(x+1) = Q(x)$.
- **Part (iii):** Use $P(0) = 16$ to evaluate the constant from Part (ii).

Solution 2.21

(i) Let $x = 3$:

$$0 = 2P(3) \implies P(3) = 0.$$

Let $x = 1$:

$$-2P(2) = 0 \implies P(2) = 0.$$

The two roots are

$$\boxed{x = 2, \qquad x = 3.}$$

(ii) By the factor theorem, write

$$P(x) = (x-2)(x-3)Q(x).$$

Substituting into the original equation,

$$(x-3)[(x-1)(x-2)Q(x+1)] = (x-1)[(x-2)(x-3)Q(x)].$$

Dividing by $(x-1)(x-2)(x-3)$ for $x \notin \{1,2,3\}$ yields

$$Q(x+1) = Q(x).$$

Since $Q(x)$ is a polynomial, the identity $Q(x+1) = Q(x)$ forces Q to be constant. Let $Q(x) = C$.

(iii) From Part (ii),

$$P(x) = C(x-2)(x-3).$$

Given $P(0) = 16$,

$$C(-2)(-3) = 16 \implies 6C = 16 \implies C = \frac{8}{3}.$$

Therefore

$$\boxed{P(x) = \frac{8}{3}(x-2)(x-3) = \frac{8}{3}x^2 - \frac{40}{3}x + 16.}$$

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Takeaways 2.21

- **Strategic substitution:** Annihilating factors is the fastest way to extract roots from functional equations.
- **The factor theorem:** Converting roots into linear factors isolates the unknown components of the polynomial.
- **Periodic polynomials:** The condition $Q(x+1) = Q(x)$ is a standard proof that a polynomial is a constant.

2.3 Advanced

Problem 2.22: A Similar Sum-of-Squares Quartic

Consider the polynomial
$$P(x) = x^4 - 7x^2 + 6x + 25.$$

(i) Show that $P(x)$ can be expressed in the form $(x^2-a)^2 + (x-b)^2$.

(ii) How many x-intercepts does the graph of $P(x)$ have?

Hint 2.22

Expand $(x^2-a)^2 + (x-b)^2$ and compare coefficients.

Solution 2.22

Expand:
$$(x^2-a)^2 + (x-b)^2 = x^4 + (-2a+1)x^2 - 2bx + (a^2+b^2).$$
Comparing with $x^4 - 7x^2 + 6x + 25$ gives
$$-2a+1 = -7, \qquad -2b = 6.$$
So
$$a = 4, \qquad b = -3.$$
Also
$$a^2 + b^2 = 16 + 9 = 25,$$
so
$$P(x) = (x^2-4)^2 + (x+3)^2.$$
As a sum of two squares, $P(x) \geq 0$ for all x, and equality would require
$$x^2 - 4 = 0 \quad \text{and} \quad x + 3 = 0$$
at the same time, which is impossible. Thus $P(x) > 0$ for all x. Hence the graph has
$$\boxed{\text{0 x-intercepts.}}$$

Takeaways 2.22

- Writing a polynomial as a sum of squares is a strong non-negativity argument.
- If the sum of squares can never be zero, the graph never meets the x-axis.

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Problem 2.23: Intersections of $(x+1)^n$ and $(x+1)^m$

At what points do the graphs of

$$f(x) = (x+1)^n, \qquad g(x) = (x+1)^m$$

intersect, where $m \neq n$?

Hint 2.23

Let $t = x+1$ and solve $t^n = t^m$.

Solution 2.23

Set $t = x+1$. Then intersections satisfy

$$t^n = t^m.$$

If $t = 0$, then both sides are 0, giving

$$x = -1 \Rightarrow (-1, 0).$$

If $t \neq 0$, then

$$t^{n-m} = 1.$$

Over the reals this gives $t = 1$, and sometimes $t = -1$.

- $t = 1 \Rightarrow x = 0$, so $(0, 1)$ is always an intersection.
- $t = -1 \Rightarrow x = -2$. This works exactly when $(-1)^n = (-1)^m$, i.e. when m and n have the same parity.

Hence the graphs always intersect at

$$\boxed{(-1, 0) \text{ and } (0, 1),}$$

and they also intersect at $x = -2$ when m and n have the same parity. The corresponding point is $(-2, 1)$ if both are even, and $(-2, -1)$ if both are odd.

Takeaways 2.23

- The substitution $t = x+1$ exposes the real structure immediately.
- In power equations, always check $t = 0$, 1, and -1.

Problem 2.24: A Factor-Theorem Identity in Three Variables

Use the factor theorem to prove that

$$x + y + z$$

is a factor of

$$x^3 + y^3 + z^3 - 3xyz.$$

Hence find the other factor.

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Hint 2.24

Treat the expression as a polynomial in x and substitute $x = -(y+z)$.

Solution 2.24

Let

$$F(x) = x^3 + y^3 + z^3 - 3xyz,$$

where y and z are constants. Then

$$\begin{aligned} F(-(y+z)) &= -(y+z)^3 + y^3 + z^3 + 3yz(y+z) \\ &= 0. \end{aligned}$$

So $x+y+z$ is a factor.
The full factorisation is

$$x^3 + y^3 + z^3 - 3xyz = (x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx).$$

Hence the other factor is

$$\boxed{x^2 + y^2 + z^2 - xy - yz - zx.}$$

Takeaways 2.24

- The factor theorem also works when the coefficients involve other variables.
- This is a standard factorisation worth remembering.

Problem 2.25: A Common Quadratic Factor

If a and b are non-zero and $a + b = 0$, prove that

$$A(x) = x^3 + ax^2 - x + b$$

and

$$B(x) = x^3 + bx^2 - x + a$$

have a common factor of degree 2 but are not identical. What is the common factor?

Hint 2.25

Replace b by $-a$ and factor by grouping.

Solution 2.25

Since $a + b = 0$, we have $b = -a$. Then

$$A(x) = x^3 + ax^2 - x - a = x^2(x+a) - 1(x+a) = (x+a)(x^2-1),$$

and

$$B(x) = x^3 - ax^2 - x + a = x^2(x-a) - 1(x-a) = (x-a)(x^2-1).$$

Therefore the common quadratic factor is

$$\boxed{x^2 - 1.}$$

They are not identical because $a \neq 0$, so their linear factors are different.

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Takeaways 2.25

- A condition like $a + b = 0$ should be used immediately.
- Grouping is often enough to reveal a hidden common factor.

Problem 2.26: Intersections via $P(x) - Q(x)$

By factoring
$$F(x) = P(x) - Q(x),$$
describe the intersections between the curves
$$P(x) = x^4 + 2x^3 + 4x - 1$$
and
$$Q(x) = x^4 + 3x^2 + 5x - 4.$$

Hint 2.26

The curves intersect exactly when $P(x) - Q(x) = 0$.

Solution 2.26

$$F(x) = P(x) - Q(x) = 2x^3 - 3x^2 - x + 3.$$
Factor by grouping:
$$2x^3 - 3x^2 - x + 3 = x^2(2x - 3) - 1(2x - 3) = (x^2 - 1)(2x - 3).$$
So the intersection x-coordinates are
$$x = -1, \quad x = 1, \quad x = \frac{3}{2}.$$
Now
$$P(-1) = Q(-1) = -4, \qquad P(1) = Q(1) = 6, \qquad P\left(\frac{3}{2}\right) = Q\left(\frac{3}{2}\right) = \frac{77}{16}.$$
Hence the curves intersect at
$$\boxed{(-1, -4),\ (1, 6)\,,\ \left(\frac{3}{2}, \frac{77}{16}\right)}\,.$$

Takeaways 2.26

- To compare two curves, subtract first and solve one equation.
- The factorisation of the difference controls the number of intersections.

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Problem 2.27: The Reversal Transformation

Consider the cubic equation

$$x^3 + ax^2 + bx + c = 0,$$

where $c \neq 0$, with roots α, β, γ.

(i) By using the substitution $y = \frac{1}{x}$, find a cubic equation in y whose roots are $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$.

(ii) Hence, or otherwise, find the roots of

$$cx^3 - bx^2 + ax - 1 = 0$$

in terms of α, β, γ.

Hint 2.27

First substitute $x = \frac{1}{y}$. Then compare your new equation with the target equation by replacing the variable with its negative.

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Solution 2.27

(i) Let

$$y = \frac{1}{x},$$

so

$$x = \frac{1}{y}.$$

Substitute into

$$x^3 + ax^2 + bx + c = 0:$$

$$\left(\frac{1}{y}\right)^3 + a\left(\frac{1}{y}\right)^2 + b\left(\frac{1}{y}\right) + c = 0.$$

Thus

$$\frac{1}{y^3} + \frac{a}{y^2} + \frac{b}{y} + c = 0.$$

Multiply by y^3:

$$1 + ay + by^2 + cy^3 = 0.$$

So the required cubic is

$$\boxed{cy^3 + by^2 + ay + 1 = 0,}$$

whose roots are

$$\boxed{\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}.}$$

(ii) From part (i), the equation

$$cx^3 + bx^2 + ax + 1 = 0$$

has roots

$$\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}.$$

Now replace x by $-z$:

$$c(-z)^3 + b(-z)^2 + a(-z) + 1 = 0.$$

This gives

$$-cz^3 + bz^2 - az + 1 = 0.$$

Multiply by -1:

$$cz^3 - bz^2 + az - 1 = 0.$$

So the roots of the target equation are the negatives of the reciprocal roots. Therefore the roots are

$$\boxed{-\frac{1}{\alpha}, -\frac{1}{\beta}, -\frac{1}{\gamma}.}$$

Takeaways 2.27

- The substitution $x = \frac{1}{y}$ reverses the order of the coefficients.
- The substitution $x = -y$ flips the signs of the odd-power terms.
- When a new polynomial looks like a reversed or sign-altered version of an old one, a substitution is usually the intended method.

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Problem 2.28: The Cosine-Polynomial Bridge

Use the identity

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta,$$

but do not prove it.
Start with

$$\cos(3\theta) = \frac{1}{2}.$$

Rewrite this equation using only $\cos\theta$. Then let

$$x = \cos\theta$$

and show that the equation becomes

$$8x^3 - 6x - 1 = 0.$$

Hence show that:

(i) $\cos\frac{\pi}{9} = \cos\frac{2\pi}{9} + \cos\frac{4\pi}{9}$

(ii) $\sec\frac{\pi}{9}\sec\frac{2\pi}{9}\sec\frac{4\pi}{9} = 8$

(iii) $\sec^2\frac{\pi}{9} + \sec^2\frac{2\pi}{9} + \sec^2\frac{4\pi}{9} = 36$

Hint 2.28

Solve $\cos(3\theta) = \frac{1}{2}$ first. Then rewrite obtuse-angle cosines using $\cos(\pi - \alpha) = -\cos\alpha$ before applying Vieta.

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Solution 2.28

From

$$\cos(3\theta) = 4\cos^3\theta - 3\cos\theta = \frac{1}{2},$$

we get

$$8\cos^3\theta - 6\cos\theta - 1 = 0.$$

Let $x = \cos\theta$. Then

$$\boxed{8x^3 - 6x - 1 = 0.}$$

Now $\cos(3\theta) = \frac{1}{2}$ gives

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3},$$

so the roots are

$$\cos\frac{\pi}{9},\ \cos\frac{5\pi}{9},\ \cos\frac{7\pi}{9} = \cos\frac{\pi}{9},\ -\cos\frac{4\pi}{9},\ -\cos\frac{2\pi}{9}.$$

(i) Since the x^2 coefficient is 0,

$$\cos\frac{\pi}{9} - \cos\frac{4\pi}{9} - \cos\frac{2\pi}{9} = 0,$$

hence

$$\boxed{\cos\frac{\pi}{9} = \cos\frac{2\pi}{9} + \cos\frac{4\pi}{9}.}$$

(ii) The product of the roots is

$$-\frac{-1}{8} = \frac{1}{8},$$

so

$$\cos\frac{\pi}{9}\cos\frac{2\pi}{9}\cos\frac{4\pi}{9} = \frac{1}{8}.$$

Taking reciprocals,

$$\boxed{\sec\frac{\pi}{9}\sec\frac{2\pi}{9}\sec\frac{4\pi}{9} = 8.}$$

(iii) For reciprocal roots, let $y = \frac{1}{x}$. Then

$$8\left(\frac{1}{y}\right)^3 - 6\left(\frac{1}{y}\right) - 1 = 0 \quad\Rightarrow\quad y^3 + 6y^2 - 8 = 0.$$

Its roots are

$$\sec\frac{\pi}{9},\ -\sec\frac{4\pi}{9},\ -\sec\frac{2\pi}{9}.$$

So

$$\sum y = -6, \qquad \sum y_i y_j = 0,$$

and therefore

$$\sum y^2 = (\sum y)^2 - 2\sum y_i y_j = 36.$$

Hence

$$\boxed{\sec^2\frac{\pi}{9} + \sec^2\frac{2\pi}{9} + \sec^2\frac{4\pi}{9} = 36.}$$

Takeaways 2.28

- A multiple-angle identity can turn a trigonometric equation into a polynomial equation.
- Once the trigonometric values are identified as roots, Vieta can produce elegant identities quickly.
- Reciprocal-root substitutions are especially useful for secant and cosecant expressions.

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Problem 2.29: The Tangent Cubic

It is known that

$$\tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}.$$

(i) By letting $x = \tan\theta$, solve the equation

$$x^3 - 3x^2 - 3x + 1 = 0.$$

(ii) Hence find the exact value of

$$\tan\frac{\pi}{12} + \tan\frac{5\pi}{12}.$$

Hint 2.29

Rewrite the cubic so that the numerator and denominator in the $\tan 3\theta$ identity appear naturally. Then use Vieta for part (ii).

Solution 2.29

(i) Let

$$x = \tan\theta.$$

Then

$$\tan^3\theta - 3\tan^2\theta - 3\tan\theta + 1 = 0.$$

Rearrange:

$$3\tan\theta - \tan^3\theta = 1 - 3\tan^2\theta.$$

So

$$\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} = 1,$$

hence

$$\tan 3\theta = 1.$$

Take three distinct solutions:

$$3\theta = \frac{\pi}{4},\ \frac{5\pi}{4},\ \frac{9\pi}{4}.$$

Thus

$$\theta = \frac{\pi}{12},\ \frac{5\pi}{12},\ \frac{3\pi}{4}.$$

Therefore the roots are

$$\boxed{\tan\frac{\pi}{12},\ \tan\frac{5\pi}{12},\ \tan\frac{3\pi}{4}}$$

that is,

$$\boxed{\tan\frac{\pi}{12},\ \tan\frac{5\pi}{12},\ -1.}$$

(ii) By Vieta, the sum of the roots of

$$x^3 - 3x^2 - 3x + 1 = 0$$

is

$$-\frac{-3}{1} = 3.$$

So

$$\tan\frac{\pi}{12} + \tan\frac{5\pi}{12} + (-1) = 3.$$

Hence

$$\boxed{\tan\frac{\pi}{12} + \tan\frac{5\pi}{12} = 4.}$$

Takeaways 2.29

- Some cubics are designed to match a trigonometric identity after a smart substitution.
- Once the roots are known in trigonometric form, Vieta can evaluate sums very quickly.
- A simple root like -1 is a useful check that the transformed solutions make sense.

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Problem 2.30: The Fibonacci Polynomial Remainder

Consider the quadratic polynomial

$$P(x) = x^2 - x - 1.$$

Let α and β be the roots of $P(x) = 0$, where α is the positive root.

(i) Using the Polynomial Division Algorithm, or otherwise, explain why for any integer $n \geq 1$, we can write

$$x^n = Q_n(x)(x^2 - x - 1) + (F_n x + G_n),$$

where $Q_n(x)$ is a polynomial and F_n, G_n are integers.

(ii) By substituting $x = \alpha$, show that

$$\alpha^n = F_n \alpha + G_n.$$

(iii) Multiply both sides of the equation in part (ii) by α. Using $\alpha^2 = \alpha + 1$, show that

$$\alpha^{n+1} = (F_n + G_n)\alpha + F_n.$$

(iv) Given that F_n and G_n are integers, and $\alpha = \frac{1+\sqrt{5}}{2}$ is irrational, deduce that

$$F_{n+1} = F_n + G_n, \qquad G_{n+1} = F_n.$$

Hence prove that

$$F_{n+1} = F_n + F_{n-1}.$$

(v) Find the remainder when x^{2026} is divided by $x^2 - x - 1$.

Hint 2.30

Treat $x^2 = x + 1$ as a reduction rule. For part (iv), compare two expressions of the form $A\alpha + B$ and use the irrationality of α.

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Solution 2.30

(i) Dividing x^n by the quadratic $x^2 - x - 1$, the remainder must have degree less than 2. So
$$x^n = Q_n(x)(x^2 - x - 1) + (F_n x + G_n)$$
for some polynomial $Q_n(x)$ and constants F_n, G_n.
(ii) Since α is a root of $x^2 - x - 1 = 0$, substituting $x = \alpha$ gives
$$\alpha^n = Q_n(\alpha) \cdot 0 + F_n\alpha + G_n = F_n\alpha + G_n.$$
(iii) Multiply by α:
$$\alpha^{n+1} = F_n\alpha^2 + G_n\alpha.$$
Using $\alpha^2 = \alpha + 1$,
$$\alpha^{n+1} = F_n(\alpha + 1) + G_n\alpha = (F_n + G_n)\alpha + F_n.$$
(iv) But also
$$\alpha^{n+1} = F_{n+1}\alpha + G_{n+1}.$$
Comparing with part (iii),
$$F_{n+1}\alpha + G_{n+1} = (F_n + G_n)\alpha + F_n.$$
Since α is irrational and all coefficients are integers,
$$F_{n+1} = F_n + G_n, \qquad G_{n+1} = F_n.$$
Now replace n by $n - 1$ in the second relation:
$$G_n = F_{n-1}.$$
Hence
$$\boxed{F_{n+1} = F_n + F_{n-1}.}$$
(v) For $n = 1$,
$$x = 1 \cdot x + 0,$$
so $F_1 = 1$, $G_1 = 0$. For $n = 2$,
$$x^2 = x + 1,$$
so $F_2 = 1$, $G_2 = 1$. From part (iv),
$$F_{n+1} = F_n + F_{n-1} \quad \text{and} \quad G_n = F_{n-1}.$$
Therefore the remainder when x^{2026} is divided by $x^2 - x - 1$ is
$$\boxed{F_{2026}x + F_{2025}.}$$

Takeaways 2.30

- Dividing by a quadratic always leaves a linear remainder.
- A polynomial relation such as $x^2 = x + 1$ can reduce high powers recursively.
- The golden ratio, Fibonacci numbers, and polynomial remainders are tightly connected by the equation $x^2 - x - 1 = 0$.
- *Math is Interconnected*: The golden ratio, polynomial remainders, and the Fibonacci sequence are all just different languages describing the exact same underlying mathematical structure: $x^2 - x - 1 = 0$.
- *Irrational Independence*: A core Extension 1/2 skill is recognizing that an equation of the form $a + b\sqrt{k} = c + d\sqrt{k}$ (where a, b, c, d are rational and $\sqrt{k}$ is irrational) forces $a = c$ and $b = d$. We used this to equate the coefficients in part (iv).

Problem 2.31: The Chebyshev Recurrence

Chebyshev polynomials are arguably one of the most important families of polynomials in all of applied mathematics. While they start as a neat trigonometric trick in high school, they quickly become the backbone of modern computing and engineering.
This problem introduces the Chebyshev polynomials of the first kind, which are defined by a simple substitution from cosine functions.
Let $x = \cos\theta$ for $0 \leq \theta \leq \pi$, so $x \in [-1, 1]$. Define

$$T_n(x) = \cos(n\theta)$$

for integers $n \geq 0$.

(i) Find $T_0(x)$ and $T_1(x)$. Using the double-angle formula, show that

$$T_2(x) = 2x^2 - 1.$$

(ii) By expanding $\cos(n\theta + \theta)$ and $\cos(n\theta - \theta)$, prove that

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).$$

(iii) Use the recurrence to find $T_3(x)$ and $T_4(x)$.

(iv) **Engineering Application:** In telecommunications, a mathematical filter must stay bounded between -1 and 1 inside the allowed frequency band, where $-1 \leq x \leq 1$. But once the signal leaves this zone ($x > 1$), the polynomial should grow as steeply as possible to help block interference.

Consider your Chebyshev polynomial

$$T_3(x) = 4x^3 - 3x$$

and the standard cubic

$$S(x) = x^3.$$

Notice that both pass through $(1, 1)$. By finding $T_3'(x)$ and $S'(x)$, compare their gradients at $x = 1$. Hence show algebraically that for all $x > 1$, the Chebyshev polynomial grows faster than the standard cubic.

Hint 2.31

Add the expansions of $\cos(n\theta + \theta)$ and $\cos(n\theta - \theta)$ so the sine terms cancel. For part (iv), compare

$$T_3'(x) = 12x^2 - 3 \qquad \text{and} \qquad S'(x) = 3x^2,$$

then look at their difference for $x > 1$.

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Solution 2.31

(i) Since

$$T_0(x) = \cos 0 = 1, \qquad T_1(x) = \cos\theta = x,$$

we have

$$\boxed{T_0(x) = 1, \qquad T_1(x) = x.}$$

Also

$$T_2(x) = \cos(2\theta) = 2\cos^2\theta - 1 = 2x^2 - 1.$$

(ii) Using the sum and difference formulas,

$$\cos(n\theta + \theta) = \cos(n\theta)\cos\theta - \sin(n\theta)\sin\theta,$$

$$\cos(n\theta - \theta) = \cos(n\theta)\cos\theta + \sin(n\theta)\sin\theta.$$

Add them:

$$\cos((n+1)\theta) + \cos((n-1)\theta) = 2\cos(n\theta)\cos\theta.$$

Substitute $T_n(x) = \cos(n\theta)$ and $x = \cos\theta$:

$$T_{n+1}(x) + T_{n-1}(x) = 2xT_n(x),$$

so

$$\boxed{T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x).}$$

(iii) Apply the recurrence:

$$T_3(x) = 2x(2x^2 - 1) - x = 4x^3 - 3x,$$

$$T_4(x) = 2x(4x^3 - 3x) - (2x^2 - 1) = 8x^4 - 8x^2 + 1.$$

(iv) Left as a lovely challenge problem. It gives a nice glimpse of why Chebyshev polynomials matter so much in engineering and signal processing.

Takeaways 2.31

- The substitution $x = \cos\theta$ turns trigonometric identities into polynomial recurrences.
- The Chebyshev polynomials of the first kind are defined by

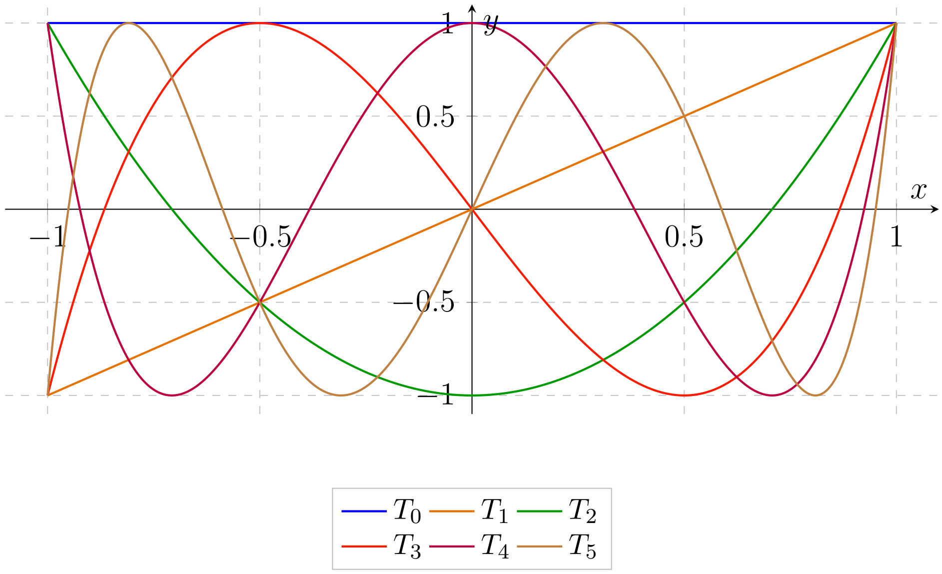
$$T_n(x) = \cos(n\theta) \qquad (x = \cos\theta).$$

- The Chebyshev polynomials of the second kind are defined by

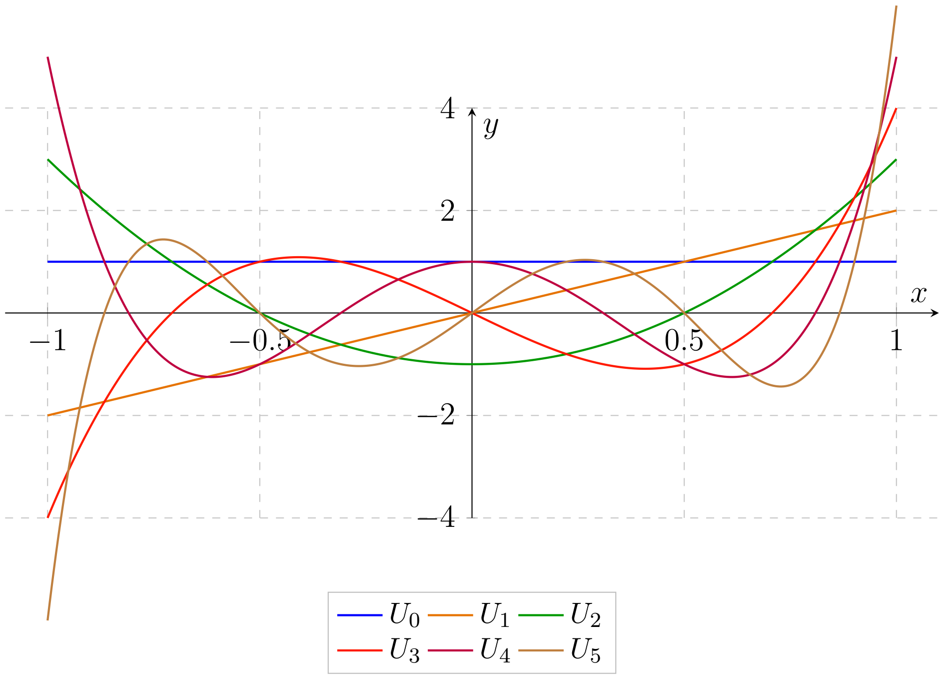
$$U_n(x) = \frac{\sin((n+1)\theta)}{\sin\theta} \qquad (x = \cos\theta).$$

- These recurrences give fast formulas for $\cos(n\theta)$ without repeated expansion.
- *Roots are Predictable*: Because these polynomials are born from a cosine wave, their roots will always be real, distinct, and trapped strictly between -1 and 1. In higher mathematics, these roots (called "Chebyshev Nodes") are the optimal points to pick when trying to curve-fit data!
- *Chebyshev Polynomials are Everywhere*: These polynomials are not just a theoretical curiosity; they play a crucial role in numerical analysis, approximation theory, and even in your 5G networks.

Chebyshev Graphs: First Kind



Chebyshev Graphs: Second Kind



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Problem 2.32: Hermite and the Bell Curve

The Gaussian function, $f(x) = e^{-x^2}$, is the mathematical foundation of the famous "Bell Curve" used in probability and statistics. By taking successive derivatives of this curve, we generate a family of functions known as the "Physicist's Hermite Polynomials" denoted as $H_n(x)$.They are defined explicitly by Rodrigues' Formula:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n}\left(e^{-x^2}\right) \quad \text{for } n \geq 0.$$

(Note: $\frac{d^0}{dx^0}$ implies no differentiation, so $H_0(x) = 1$)

(i) By manually applying the Chain Rule and Product Rule, find explicit formulas for $H_1(x)$, $H_2(x)$, and $H_3(x)$.

(ii) Find the roots of $H_3(x) = 0$. Hence state a conjecture about the parity of $H_n(x)$ when n is even or odd.

(iii) Starting from

$$(-1)^n \frac{d^n}{dx^n}\left(e^{-x^2}\right) = e^{-x^2} H_n(x),$$

prove that

$$H_{n+1}(x) = 2xH_n(x) - H_n'(x).$$

(iv) **Quantum Mechanics Application:** In quantum physics, the probability of finding a vibrating particle at position x in its nth energy state is proportional to

$$P_n(x) = e^{-x^2}[H_n(x)]^2.$$

For the second excited state ($n = 2$), use your formula for $H_2(x)$ to build $P_2(x)$, find $P_2'(x)$, and determine the exact x-coordinates of the local maxima.

Hint 2.32

Differentiate step by step in part (i). In part (iii), differentiate both sides and use the product rule carefully. Part (iv) is left as an extension challenge for readers curious about quantum mechanics.

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Solution 2.32

(i) First,

$$H_1(x) = -e^{x^2}\frac{d}{dx}(e^{-x^2}) = -e^{x^2}(-2xe^{-x^2}) = \boxed{2x}.$$

Next,

$$\frac{d^2}{dx^2}(e^{-x^2}) = \frac{d}{dx}(-2xe^{-x^2}) = -2e^{-x^2} + 4x^2e^{-x^2}.$$

So

$$H_2(x) = e^{x^2}e^{-x^2}(4x^2-2) = \boxed{4x^2-2}.$$

Then

$$\frac{d^3}{dx^3}(e^{-x^2}) = \frac{d}{dx}\big(e^{-x^2}(4x^2-2)\big) = e^{-x^2}(-8x^3+12x).$$

Hence

$$H_3(x) = -e^{x^2}e^{-x^2}(-8x^3+12x) = \boxed{8x^3-12x}.$$

(ii) Solve

$$8x^3-12x = 0 \Rightarrow 4x(2x^2-3) = 0.$$

So the roots are

$$\boxed{x = 0,\ \pm\sqrt{\frac{3}{2}}.}$$

Also H_1 and H_3 are odd, while H_0 and H_2 are even. So the natural conjecture is:

$$\boxed{H_n(x) \text{ is even if } n \text{ is even, and odd if } n \text{ is odd.}}$$

(iii) Differentiate

$$(-1)^n\frac{d^n}{dx^n}(e^{-x^2}) = e^{-x^2}H_n(x).$$

The left-hand side becomes

$$(-1)^n\frac{d^{n+1}}{dx^{n+1}}(e^{-x^2}) = -e^{-x^2}H_{n+1}(x).$$

The right-hand side gives

$$\frac{d}{dx}\big(e^{-x^2}H_n(x)\big) = (-2xe^{-x^2})H_n(x) + e^{-x^2}H_n'(x).$$

So

$$-e^{-x^2}H_{n+1}(x) = e^{-x^2}\big(-2xH_n(x) + H_n'(x)\big).$$

Divide by $-e^{-x^2}$:

$$\boxed{H_{n+1}(x) = 2xH_n(x) - H_n'(x).}$$

(iv) Left as a challenge problem. This is a lovely invitation to explore quantum mechanics, where Hermite polynomials describe the quantum harmonic oscillator.

Takeaways 2.32

- A good recurrence can replace repeated heavy differentiation.
- Hermite polynomials alternate parity: even index gives an even polynomial, odd index gives an odd polynomial.
- Rodrigues' formula is a powerful bridge between calculus and polynomial structure.
- Hermite polynomials also appear naturally in quantum mechanics.

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Problem 2.33: Bernoulli Polynomials and the Beta Distribution

The Bernoulli polynomials, $B_n(x)$, were discovered by Swiss mathematician Jacob Bernoulli in the early 1700s (published posthumously in his masterwork Ars Conjectandi). Originally developed to solve centuries-old number theory puzzles, these polynomials have become surprisingly crucial in modern applied mathematics. In higher education, they form the foundation of the Euler-Maclaurin formula (used in numerical analysis to approximate complex integrals) and frequently appear in advanced statistics to define the moments, cumulants, and error bounds of Probability Density Functions (PDFs).

They are defined by a starting seed $B_0(x) = 1$, and two strict calculus rules for all integers $n \geq 1$:

(1) The Derivative Rule:

$$\frac{d}{dx} B_n(x) = nB_{n-1}(x)$$

(2) The Integral Rule:

$$\int_0^1 B_n(x)\, dx = 0$$

Given that

$$B_0(x) = 1, \quad B_1(x) = x - \frac{1}{2}, \quad \text{and} \quad B_2(x) = x^2 - x + \frac{1}{6}$$

(i) Use the Derivative and Integral rules to find the explicit algebraic expressions for $B_3(x)$ and $B_4(x)$.

(ii) A remarkable property of Bernoulli polynomials is their symmetry. Verify algebraically that $B_3(1 - x) = -B_3(x)$. Hence, without further calculation, state the three roots of the cubic equation $B_3(x) = 0$.

(iii) In probability theory, a valid Probability Density Function (PDF) defined on the interval $0 \leq x \leq 1$ must satisfy two conditions: $f(x) \geq 0$ for all x, and $\int_0^1 f(x)\, dx = 1$. Consider the function $f(x) = k\left(B_4(x) + \frac{1}{30}\right)$. Use the Integral Rule to quickly determine the value of the constant k. Then, use your polynomial $B_4(x)$ from part (i) to factorise $f(x)$ and explicitly justify why $f(x) \geq 0$ on this domain.

(iv) The distribution defined in part (iii) is a special case of the Beta Distribution, a continuous probability distribution widely used in higher-level statistics to model random proportions (such as the reliability of a mechanical system or the conversion rate of a website). Using your factorised form of $f(x)$, calculate the Expected Value, $E(X)$, of this distribution.

Hint 2.33

Integrate to get B_3 and B_4, then use the condition $\int_0^1 B_n(x)\, dx = 0$ to find the constants. In part (iii), do not integrate B_4 directly: use the rule you were given.

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Solution 2.33

(i) Since

$$B_3'(x) = 3B_2(x) = 3x^2 - 3x + \frac{1}{2},$$

integration gives

$$B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x + C.$$

Using $\int_0^1 B_3(x)\,dx = 0$ gives $C = 0$, so

$$\boxed{B_3(x) = x^3 - \frac{3}{2}x^2 + \frac{1}{2}x.}$$

Next,

$$B_4'(x) = 4B_3(x) = 4x^3 - 6x^2 + 2x,$$

so

$$B_4(x) = x^4 - 2x^3 + x^2 + D.$$

Using $\int_0^1 B_4(x)\,dx = 0$ gives $D = -\frac{1}{30}$, hence

$$\boxed{B_4(x) = x^4 - 2x^3 + x^2 - \frac{1}{30}.}$$

(ii) Substitute $1 - x$ into B_3:

$$B_3(1-x) = -(x^3 - \tfrac{3}{2}x^2 + \tfrac{1}{2}x) = -B_3(x).$$

Also

$$B_3(x) = x\left(x^2 - \frac{3}{2}x + \frac{1}{2}\right) = x(x-1)\left(x - \frac{1}{2}\right),$$

so the roots are

$$\boxed{0,\ \frac{1}{2},\ 1.}$$

(iii) Since $\int_0^1 f(x)\,dx = 1$,

$$k\int_0^1 \left(B_4(x) + \frac{1}{30}\right)dx = 1.$$

Using $\int_0^1 B_4(x)\,dx = 0$,

$$\frac{k}{30} = 1 \Rightarrow \boxed{k = 30.}$$

Now

$$f(x) = 30\left(x^4 - 2x^3 + x^2\right) = 30x^2(x-1)^2.$$

This is non-negative on $[0, 1]$, since it is a product of squares.

(iv)

$$E(X) = \int_0^1 x f(x)\,dx = 30\int_0^1 x^3(x-1)^2\,dx.$$

So

$$E(X) = 30\int_0^1 (x^5 - 2x^4 + x^3)\,dx = 30\left(\frac{1}{6} - \frac{2}{5} + \frac{1}{4}\right) = \boxed{\frac{1}{2}.}$$

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Takeaways 2.33

- The derivative and zero-integral rules determine Bernoulli polynomials recursively.
- Symmetry can reveal roots quickly.
- *Beta Distributions* is a family of continuous probability distributions defined on the interval $[0, 1]$. They are parameterized by two positive shape parameters, α and β, which determine the shape of the distribution. The Beta distribution is widely used in statistics to model random variables that represent proportions or probabilities, such as the success rate of a process or the proportion of a population with a certain characteristic. It is defined by the probability density function:

  $$f(x; \alpha, \beta) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}$$

  where $B(\alpha, \beta)$ is the Beta function, which serves as a normalization constant to ensure that the total area under the curve is 1 and is given by:

  $$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt.$$

- *Beta Distributions in the Real World*: The function $f(x) = 30x^2(1-x)^2$ is formally known as a Beta$(3, 3)$ distribution. Because its domain is strictly $[0, 1]$, university statisticians use it to model percentages and probabilities, such as predicting the batting average of a baseball player or the defect rate in a manufacturing plant!
- *Interconnected Math*: This problem beautifully demonstrates that pure calculus (polynomial derivation) and applied statistics (Probability Density Functions) are not isolated topics. The algebraic properties of a polynomial directly dictate its viability as a statistical model.

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Problem 2.34: Legendre Polynomials and the Infinite Dot Product

In the study of vectors, two vectors are defined as orthogonal (perpendicular) if their dot product is zero: $\vec{u} \cdot \vec{v} = 0$.
In advanced calculus, we can apply this exact same concept to functions. We define the "inner product" of two functions $f(x)$ and $g(x)$ over the interval $[-1, 1]$ as the definite integral of their product. Thus, two functions are **orthogonal** if:

$$\int_{-1}^{1} f(x)g(x)\,dx = 0.$$

The **Legendre Polynomials**, denoted $P_n(x)$, form a sequence of orthogonal polynomials. The first three are:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1).$$

(i) Verify that P_0 and P_1 are orthogonal, and that P_1 and P_2 are orthogonal.

(ii) Let $Q(x) = ax^2 + bx + c$. Show that if Q is orthogonal to both P_0 and P_1, then Q is a scalar multiple of P_2.

(iii) Using

$$(n+1)P_{n+1}(x) = (2n+1)xP_n(x) - nP_{n-1}(x),$$

(do NOT prove this) find $P_3(x)$.

(iv) **Application to Numerical Analysis:**

In computer science, numerical integration usually provides an estimate rather than a perfect answer. The two-point Gaussian Quadrature rule estimates the integral of a general function $g(x)$ by sampling it at the roots of the Legendre polynomial $P_2(x)$:

$$\int_{-1}^{1} g(x)\,dx \approx g(\alpha) + g(\beta),$$

where α and β are the roots of $P_2(x) = 0$.

Remarkably, the theorem guarantees that this is not just an approximation, but an exact equality for any polynomial of degree 3 or less.

Verify that for every cubic

$$f(x) = ax^3 + bx^2 + cx + d,$$

the formula is exact:

$$\int_{-1}^{1} f(x)\,dx = f(\alpha) + f(\beta).$$

Hint 2.34

For part (ii), write down the two orthogonality integrals and simplify. For part (iv), first solve $P_2(x) = 0$, then use the symmetry $\beta = -\alpha$ to simplify the odd-power terms.

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Solution 2.34

(i)

$$\int_{-1}^{1} P_0(x)P_1(x)\,dx = \int_{-1}^{1} x\,dx = 0.$$

Also

$$\int_{-1}^{1} P_1(x)P_2(x)\,dx = \frac{1}{2}\int_{-1}^{1}(3x^3 - x)\,dx = 0,$$

since the integrand is odd.

(ii) Let $Q(x) = ax^2 + bx + c$. Orthogonal to P_0:

$$\int_{-1}^{1}(ax^2 + bx + c)\,dx = \frac{2a}{3} + 2c = 0 \Rightarrow c = -\frac{a}{3}.$$

Orthogonal to P_1:

$$\int_{-1}^{1}(ax^2 + bx + c)x\,dx = \int_{-1}^{1}(ax^3 + bx^2 + cx)\,dx = \frac{2b}{3} = 0,$$

so $b = 0$. Hence

$$Q(x) = ax^2 - \frac{a}{3} = \frac{a}{3}(3x^2 - 1) = \frac{2a}{3}P_2(x).$$

So Q is a scalar multiple of P_2.

(iii) Put $n = 2$:

$$3P_3(x) = 5xP_2(x) - 2P_1(x).$$

Thus

$$3P_3(x) = 5x \cdot \frac{1}{2}(3x^2 - 1) - 2x = \frac{15x^3 - 9x}{2},$$

so

$$\boxed{P_3(x) = \frac{1}{2}(5x^3 - 3x).}$$

(iv) The roots of $P_2(x) = \frac{1}{2}(3x^2 - 1)$ are $\alpha = \frac{1}{\sqrt{3}}, \qquad \beta = -\frac{1}{\sqrt{3}}$. Now

$$\int_{-1}^{1}(ax^3 + bx^2 + cx + d)\,dx = \frac{2b}{3} + 2d,$$

while

$$f(\alpha) + f(\beta) = a(\alpha^3 + \beta^3) + b(\alpha^2 + \beta^2) + c(\alpha + \beta) + 2d = \frac{2b}{3} + 2d,$$

because $\beta = -\alpha$, so the odd-power terms cancel and

$$\alpha^2 + \beta^2 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}.$$

Hence

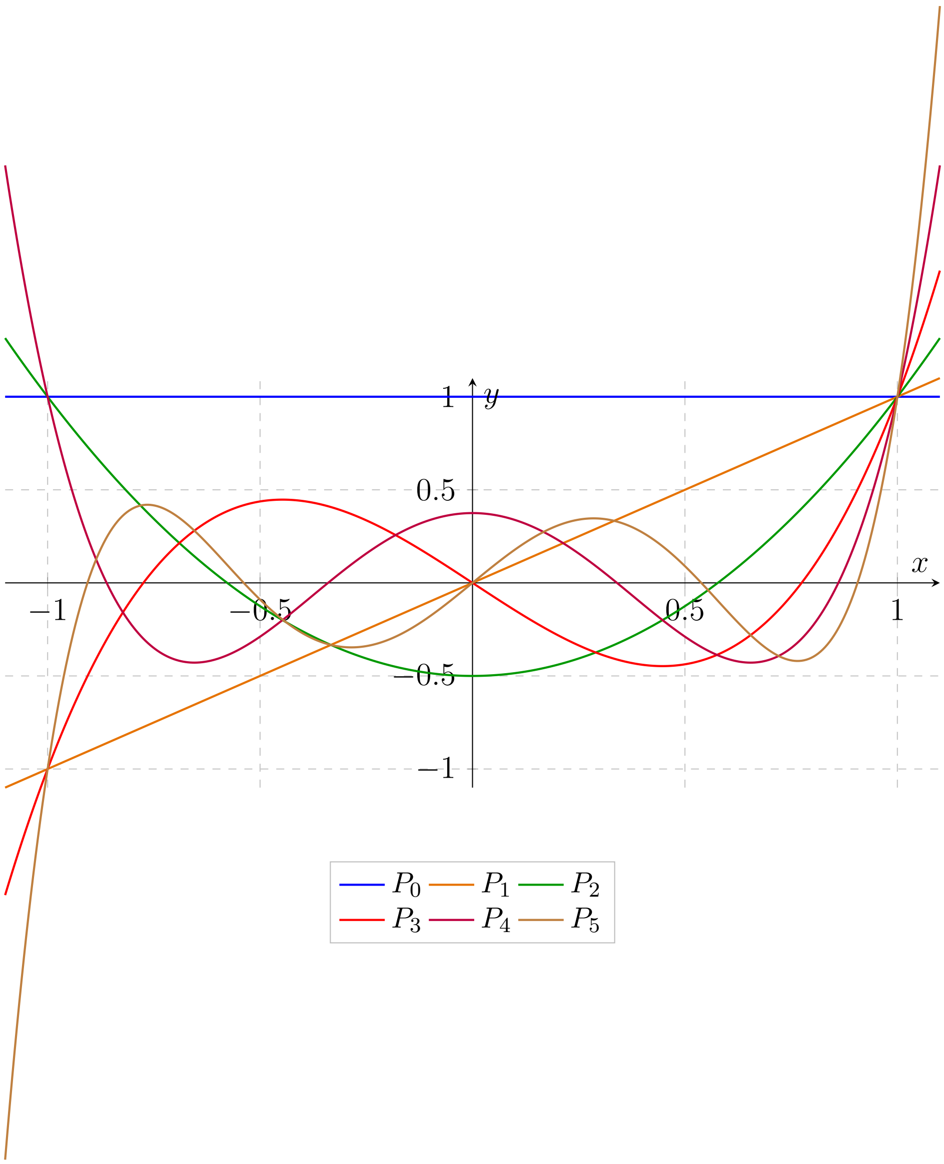
$$\boxed{\int_{-1}^{1} f(x)\,dx = f(\alpha) + f(\beta).}$$

This is a beautiful example of Gaussian quadrature. The general theory needs more heavy lifting and quite a bit of manual calculation.

Takeaways 2.34

- Orthogonality of functions is the continuous version of a dot product being zero.
- Symmetry on $[-1, 1]$ is a powerful shortcut: odd integrands integrate to zero.
- Legendre, Chebyshev, and Hermite polynomials are all examples of orthogonal polynomial families.
- The roots of orthogonal polynomials also power Gaussian quadrature, one of the most important ideas in numerical integration.
- *Calculus Meets Vectors*: In the HSC syllabus, you learn the dot product as $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3 = 0$. An integral is simply an infinite sum of tiny slices. Therefore, $\int f(x)g(x)\,dx = 0$ is literally the dot product of two continuous functions with infinitely many dimensions!

Legendre Graphs



Problem 2.35: Telescoping Polynomials and Trigonometry

Consider the polynomial $P(x) = 4x^4 - 9x^2 + 2x + 8$.

(i) Find a quadratic polynomial $f(x) = ax^2 + bx$ and a constant C such that

$$P(x) = f(2x^2 - 1) - f(x) + C.$$

(ii) By substituting $x = \cos\theta$, deduce that

$$P(\cos\theta) = f(\cos 2\theta) - f(\cos\theta) + C.$$

(iii) Let $\theta_k = \dfrac{2^k\pi}{1023}$. Write out $\displaystyle\sum_{k=1}^{10} P(\cos\theta_k)$ using part (ii), and show it simplifies to

$$f(\cos\theta_{11}) - f(\cos\theta_1) + 10C.$$

(iv) Hence, evaluate the exact value of the sum

$$\sum_{k=1}^{10} P\left(\cos\frac{2^k\pi}{1023}\right).$$

Hint 2.35

- **(i):** Expand $f(2x^2 - 1) - f(x) + C$ and equate coefficients to $P(x)$.
- **(iii):** Notice $\theta_{k+1} = 2\theta_k$. Write the first few terms to observe the sum telescoping.
- **(iv):** Calculate θ_{11}. Since cosine has period 2π, relate $\cos(\theta_{11})$ back to $\cos(\theta_1)$.

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Solution 2.35

(i) The Difference Polynomial
Let $f(x) = ax^2 + bx$. Expanding $f(2x^2 - 1) - f(x) + C$ yields:

$$a(2x^2-1)^2 + b(2x^2-1) - ax^2 - bx + C = 4ax^4 + (2b-5a)x^2 - bx + (a - b + C).$$

Equating coefficients with $P(x) = 4x^4 - 9x^2 + 2x + 8$:

- x^4: $4a = 4 \implies \mathbf{a = 1}$.
- x^1: $-b = 2 \implies \mathbf{b = -2}$. (Check x^2: $2(-2) - 5(1) = -9$ ✓)
- Constant: $1 - (-2) + C = 8 \implies \mathbf{C = 5}$.

Thus $\boxed{f(x) = x^2 - 2x}$ and $\boxed{C = 5}$.

(ii) Trigonometric Substitution
Let $x = \cos\theta$. The double-angle identity $2\cos^2\theta - 1 = \cos(2\theta)$ gives $2x^2 - 1 = \cos(2\theta)$. Substituting into part (i):

$$\boxed{P(\cos\theta) = f(\cos 2\theta) - f(\cos\theta) + 5.}$$

(iii) Telescoping the Sum
Since $\theta_k = \dfrac{2^k\pi}{1023}$, we have $2\theta_k = \theta_{k+1}$. Applying part (ii):

$$\sum_{k=1}^{10} P(\cos\theta_k) = \sum_{k=1}^{10}\big[f(\cos\theta_{k+1}) - f(\cos\theta_k) + 5\big].$$

This is a telescoping series:

$$\begin{aligned} &= \big[f(\cos\theta_2) - f(\cos\theta_1) + 5\big] + \big[f(\cos\theta_3) - f(\cos\theta_2) + 5\big] + \cdots + \big[f(\cos\theta_{11}) - f(\cos\theta_{10}) + 5\big] \\ &= \boxed{f(\cos\theta_{11}) - f(\cos\theta_1) + 50.} \end{aligned}$$

(iv) Evaluating the Final Sum
Compute the final angle:

$$\theta_{11} = \frac{2^{11}\pi}{1023} = \frac{2048\pi}{1023} = \frac{(2\times 1023 + 2)\pi}{1023} = 2\pi + \frac{2\pi}{1023} = 2\pi + \theta_1.$$

Because $\cos(2\pi + \theta) = \cos\theta$, we have $\cos\theta_{11} = \cos\theta_1$. Therefore:

$$f(\cos\theta_{11}) - f(\cos\theta_1) = 0.$$

Hence the sum evaluates exactly to

$$\boxed{50.}$$

Takeaways 2.35

- **Polynomial–Trig Duality:** The expression $2x^2 - 1$ is the algebraic form of the double-angle formula $\cos(2\theta) = 2\cos^2\theta - 1$. Recognising this connection converts a polynomial question into a telescoping trigonometric sum.
- **Telescoping Collapse:** Expressing each term as a difference $f(x_{k+1}) - f(x_k)$ causes a large finite sum to instantly collapse to its two boundary values — a powerful technique whenever the index can be "stepped" by a fixed operation.
- **Periodicity as a Final Weapon:** Once the boundary term $\cos\theta_{11}$ is recognised as $\cos\theta_1$ via the period 2π, the answer reduces to $10C = 50$ with no further calculation.

3 Part 2: Practice and Challenge Bank

3.1 Warm-Up Drills

Problem 3.1: Multiplicity Table and Sketch

Let

$$P(x) = x^3(x-2)^4(x^2-x+1).$$

Discuss the behaviour of $P(x)$ around $-\infty$, $+\infty$, 0 and 2. Summarise it in a table and hence sketch the graph roughly.

Hint 3.1

Note that $x^2 - x + 1 > 0$ for all real x.

Solution 3.1

The degree is 9 and the leading coefficient is positive, so

$$x \to -\infty \Rightarrow P(x) \to -\infty, \qquad x \to +\infty \Rightarrow P(x) \to +\infty.$$

Also

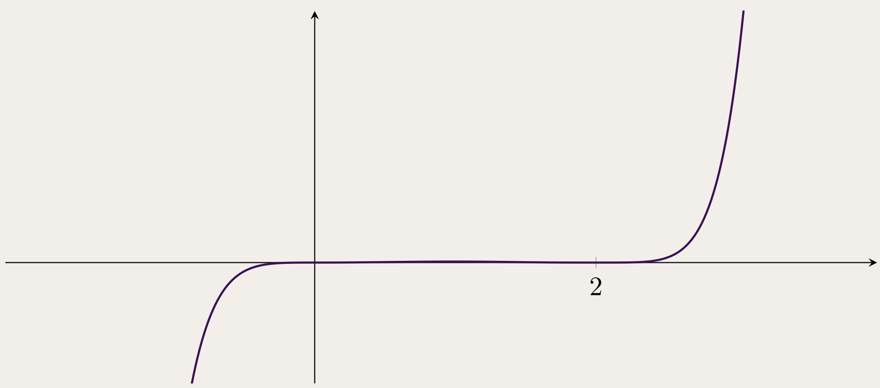
$$x^2 - x + 1 = \left(x - \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

for all real x.
So the only real zeroes are

$$x = 0 \text{ with multiplicity } 3, \qquad x = 2 \text{ with multiplicity } 4.$$

Hence

Point	Behaviour
$x \to -\infty$	$P(x) \to -\infty$
$x = 0$	crosses with flattening
$x = 2$	touches and turns
$x \to +\infty$	$P(x) \to +\infty$



Takeaways 3.1

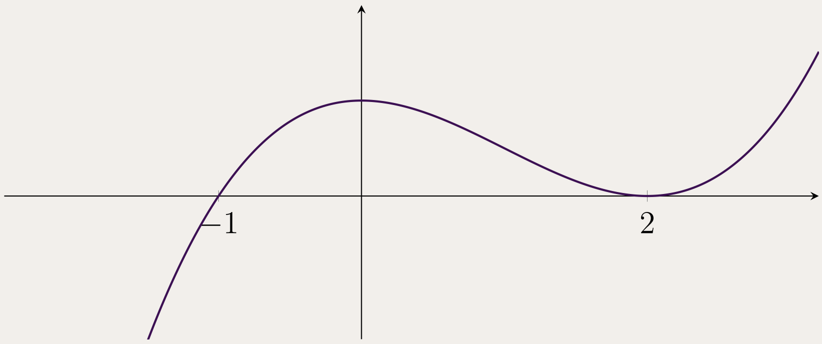
- A factor that is always positive does not change the sign pattern.
- Even very complicated sketches are manageable once the multiplicities are identified.

Problem 3.2: Reading a Cubic from Its Sketch

The graph below belongs to one of the following polynomials. Which one is correct?

(a) $(x-2)^2(x+1)$ (b) $(x-2)(x+1)^2$

(c) $(x-2)(x+1)$ (d) $(x+2)^2(x+1)$



Hint 3.2

One intercept is a crossing and the other is a touch.

Solution 3.2

The graph crosses the axis at $x=-1$, so that root has multiplicity 1. It touches and turns at $x=2$, so that root has multiplicity 2. Therefore the correct choice is

$$\boxed{\text{(a) } (x-2)^2(x+1).}$$

Takeaways 3.2

- Root multiplicity can be read directly from the graph.
- A correct factor form must match both the intercepts and the local shape.

Problem 3.3: A Quartic with Paired Root Sums

Consider the polynomial equation

$$x^4 - 8x^3 + 20x^2 + kx + 3 = 0,$$

where k is real. It is given that the sum of two of the roots equals the sum of the other two roots.

(i) Find k.

(ii) Hence find all four roots.

Hint 3.3

If each pair has the same sum, use two quadratic factors with the same x-coefficient.

Solution 3.3

Let the roots be $\alpha, \beta, \gamma, \delta$ with

$$\alpha + \beta = \gamma + \delta.$$

Since the sum of all four roots is 8, each pair sums to 4. So write

$$x^4 - 8x^3 + 20x^2 + kx + 3 = (x^2 - 4x + A)(x^2 - 4x + B).$$

Expanding,

$$x^4 - 8x^3 + (A + B + 16)x^2 - 4(A + B)x + AB.$$

Therefore

$$A + B = 4, \qquad AB = 3, \qquad k = -4(A + B) = -16.$$

So

$$\boxed{k = -16.}$$

Now $\{A, B\} = \{1, 3\}$, hence

$$x^4 - 8x^3 + 20x^2 - 16x + 3 = (x^2 - 4x + 1)(x^2 - 4x + 3).$$

Thus the roots are

$$\boxed{2 - \sqrt{3},\ 1,\ 3,\ 2 + \sqrt{3}.}$$

Takeaways 3.3

- A condition on the roots often suggests a useful factorisation pattern.
- This is a good example of Vieta plus coefficient comparison working together.

Problem 3.4: Reading the Degree from a Graph

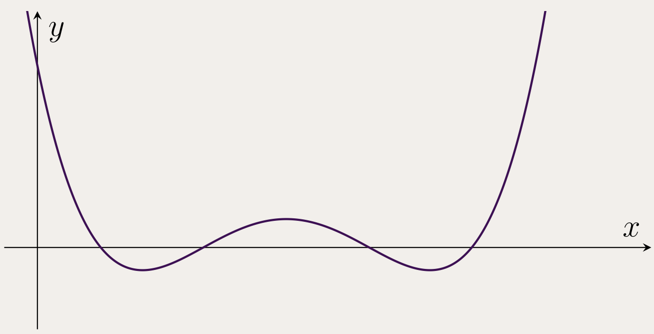
The graph below is the graph of a polynomial. What could be the degree of the polynomial? Explain your answer.

(a) 1

(b) 2

(c) 3

(d) 4



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Hint 3.4

Look at the end behaviour and count the maximum possible number of turning points.

Solution 3.4

The graph has both ends rising, so the degree is even with positive leading coefficient. Also, the curve has the shape of a quartic and can have up to 3 turning points. Among the options given, the only suitable degree is

$$\boxed{\text{(d) } 4.}$$

Takeaways 3.4

- A polynomial of degree n has at most $n-1$ turning points.
- If both ends of the graph go up, the degree must be even and the leading coefficient positive.

3.2 Stretch Problems

Problem 3.5: A Family Determined by Four Intersections

Interpret the intended question as follows: the horizontal line $y = k$ meets the quartic

$$y = ax^4 + bx^3 + cx^2 + dx + k$$

at $x = -1, 1, 2, 3$. Find b, c and d in terms of a.

Hint 3.5

The intersections with $y = k$ are the roots of $ax^4 + bx^3 + cx^2 + dx$.

Solution 3.5

Since the curve meets $y = k$ at $x = -1, 1, 2, 3$, the polynomial

$$ax^4 + bx^3 + cx^2 + dx$$

has roots $-1, 1, 2, 3$. So

$$ax^4 + bx^3 + cx^2 + dx = a(x+1)(x-1)(x-2)(x-3).$$

Now

$$(x+1)(x-1) = x^2 - 1, \qquad (x-2)(x-3) = x^2 - 5x + 6,$$

so

$$(x^2-1)(x^2-5x+6) = x^4 - 5x^3 + 5x^2 + 5x - 6.$$

Therefore

$$ax^4 + bx^3 + cx^2 + dx = a(x^4 - 5x^3 + 5x^2 + 5x - 6).$$

Comparing coefficients,

$$\boxed{b = -5a, \qquad c = 5a, \qquad d = 5a.}$$

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Takeaways 3.5

- Intersections with a horizontal line come from the shifted polynomial $P(x) - k$.
- Once the roots are known, the factorised form is immediate up to scale.

Problem 3.6: A Surd Expression from Quadratic Roots

Let x_1 and x_2 be the distinct roots of

$$x^2 - 7x + 3 = 0.$$

Calculate the exact value of

$$A = \sqrt{11x_1 + 1} + \sqrt{13x_2 + 6}.$$

Hint 3.6

For each root x, use $x^2 - 7x + 3 = 0$ to rewrite the expression under each square root as a perfect square.

Solution 3.6

Since x_1, x_2 are roots of

$$x^2 - 7x + 3 = 0,$$

Vieta gives

$$x_1 + x_2 = 7, \qquad x_1 x_2 = 3.$$

So both roots are positive.
Now use $x_i^2 - 7x_i + 3 = 0$.
For x_1:

$$11x_1 + 1 = (11x_1 + 1) + (x_1^2 - 7x_1 + 3) = x_1^2 + 4x_1 + 4 = (x_1 + 2)^2.$$

For x_2:

$$13x_2 + 6 = (13x_2 + 6) + (x_2^2 - 7x_2 + 3) = x_2^2 + 6x_2 + 9 = (x_2 + 3)^2.$$

Hence

$$A = \sqrt{(x_1 + 2)^2} + \sqrt{(x_2 + 3)^2}.$$

Because $x_1, x_2 > 0$, we have $x_1 + 2 > 0$ and $x_2 + 3 > 0$, so

$$A = (x_1 + 2) + (x_2 + 3) = x_1 + x_2 + 5 = 7 + 5 = \boxed{12}.$$

Therefore:

$$\text{exact value } = 12, \qquad \text{approximate value } = 12.0.$$

Takeaways 3.6

- Adding the root equation $x^2 - 7x + 3 = 0$ can turn linear expressions into perfect squares.
- Vieta then collapses the final expression to a simple constant.

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3.3 Challenge Corner

Problem 3.7: The Alternating Roots of e^x

Let

$$P_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}$$

be the nth Maclaurin polynomial for e^x.

(i) Assuming that for some even integer $2k$, the polynomial $P_{2k}(x) > 0$ for all real x, prove that $P_{2k+1}(x) = 0$ has exactly one real root.

(ii) Let α be the unique real root of $P_{2k+1}(x) = 0$. By finding the global minimum of $P_{2k+2}(x)$, prove that $P_{2k+2}(x) > 0$ for all real x.

(iii) Hence deduce the number of real roots of $P_n(x) = 0$ when n is even, and when n is odd.

Hint 3.7

Use

$$P_n'(x) = P_{n-1}(x)$$

and

$$P_n(x) = P_{n-1}(x) + \frac{x^n}{n!}.$$

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Solution 3.7

Use
$$P_n'(x) = P_{n-1}(x).$$
(i) If $P_{2k}(x) > 0$ for all real x, then
$$P_{2k+1}'(x) = P_{2k}(x) > 0,$$
so P_{2k+1} is strictly increasing. Since it has odd degree,
$$P_{2k+1}(x) \to -\infty \text{ as } x \to -\infty, \qquad P_{2k+1}(x) \to \infty \text{ as } x \to \infty.$$
Hence $P_{2k+1}(x) = 0$ has exactly one real root.
(ii) Let that root be α. Then
$$P_{2k+2}'(x) = P_{2k+1}(x),$$
so $x = \alpha$ is the only stationary point of P_{2k+2}, hence its global minimum. Now
$$P_{2k+2}(x) = P_{2k+1}(x) + \frac{x^{2k+2}}{(2k+2)!},$$
so
$$P_{2k+2}(\alpha) = \frac{\alpha^{2k+2}}{(2k+2)!} > 0$$
because $P_{2k+1}(\alpha) = 0$, $\alpha \neq 0$, and $2k+2$ is even. Thus
$$P_{2k+2}(x) > 0$$
for all real x.
(iii) Starting from $P_0(x) = 1 > 0$, parts (i) and (ii) give an induction chain:
$$\boxed{P_n(x) = 0 \text{ has no real roots when } n \text{ is even,}}$$
$$\boxed{P_n(x) = 0 \text{ has exactly one real root when } n \text{ is odd.}}$$

Takeaways 3.7

- The derivative relation $P_n' = P_{n-1}$ links consecutive Maclaurin polynomials.
- Evaluating the even polynomial at the odd polynomial's root makes the summation collapse to one positive term.
- This creates an induction pattern: even cases have no real roots, odd cases have exactly one.

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Problem 3.8: The Geometric Solution of the Quartic

Historical Context

Before the development of symbolic algebra in the 16th century, solving cubic and quartic equations was one of the great challenges of mathematics. While quadratic equations could be handled geometrically, higher-degree equations seemed much harder.

Omar Khayyam (1048–1131), the Persian mathematician, astronomer, and poet, gave a remarkable geometric approach: instead of searching for a symbolic formula, he represented roots as intersection points of conic sections.

The following problem recreates that idea using modern Cartesian coordinates.

Consider the rectangular hyperbola

$$\mathcal{H}: \; y = \frac{1}{x}, \qquad x \neq 0,$$

and the circle

$$\mathcal{C}: \; (x-c)^2 + y^2 = c^2,$$

where $c > 0$.

(i) By substituting the equation of the hyperbola into the equation of the circle, show that the x-coordinates of the intersection points are the roots of

$$P(x) = x^4 - 2cx^3 + 1.$$

(ii) Use $P'(x)$ to show that if

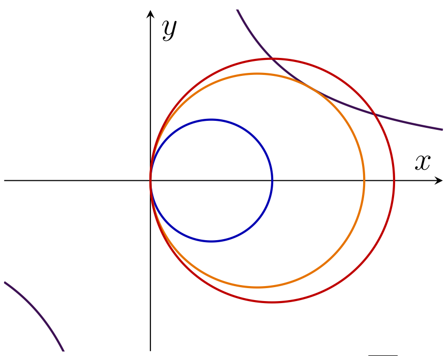
$$c > \sqrt[4]{\frac{16}{27}},$$

then $P(x)$ has exactly two distinct real roots. Interpret this geometrically.

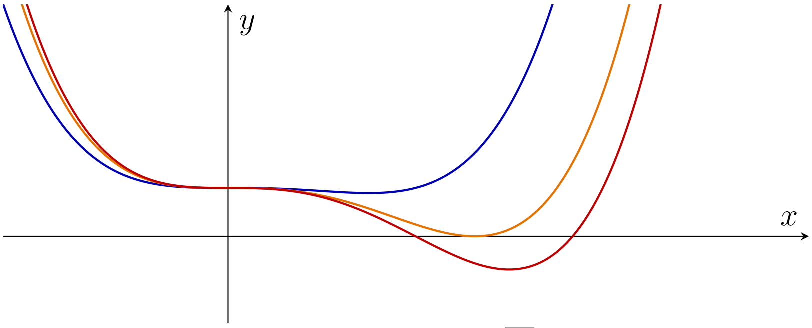
(iii) Let the four roots be $\alpha, \beta, \gamma, \delta$. Find the value of

$$\alpha\beta\gamma\delta.$$

(iv) **The Khayyam Challenge:** For what value of c does the circle become tangent to the hyperbola? Justify your answer using multiplicity of roots.



Hyperbola $\mathcal{H}$ in purple. Circles shown for $c = \frac{1}{2}$ (blue), $c = \sqrt[4]{\frac{16}{27}} \approx 0.87738$ (orange), and $c = 1$ (red).



Quartics $P(x) = x^4 - 2cx^3 + 1$ for $c = \frac{1}{2}$ (blue), $c = \sqrt[4]{\frac{16}{27}} \approx 0.87738$ (orange), and $c = 1$ (red).

Hint 3.8

For tangency, the quartic must have a repeated real root. So solve $P(x) = 0$ and $P'(x) = 0$ together.

Solution 3.8

(i) Substitute $y = \frac{1}{x}$ into the circle:

$$(x-c)^2 + \frac{1}{x^2} = c^2.$$

$$x^2 - 2cx + \frac{1}{x^2} = 0$$

and hence

$$\boxed{x^4 - 2cx^3 + 1 = 0.}$$

(ii) For

$$P(x) = x^4 - 2cx^3 + 1,$$

$$P'(x) = 4x^3 - 6cx^2 = 2x^2(2x - 3c).$$

So the relevant stationary point is $x = \frac{3c}{2}$. Also

$$P\left(\frac{3c}{2}\right) = \left(\frac{3c}{2}\right)^4 - 2c\left(\frac{3c}{2}\right)^3 + 1 = 1 - \frac{27c^4}{16}.$$

$$\text{If } c > \sqrt[4]{\frac{16}{27}}, \text{ then } P\left(\frac{3c}{2}\right) < 0.$$

Since $P(x) \to +\infty$ as $x \to \pm\infty$, the quartic crosses the x-axis exactly twice. Therefore it has exactly two distinct real roots, so geometrically the circle and hyperbola meet at exactly two real points.

(iii) By Vieta,

$$\alpha\beta\gamma\delta = \frac{1}{1} = \boxed{1}.$$

(iv) Tangency occurs when the quartic has a repeated real root. So solve

$$P(x) = 0, \qquad P'(x) = 0.$$

From $P'(x) = 0$, the repeated root must occur at $x = \frac{3c}{2}$. So

$$P\left(\frac{3c}{2}\right) = 0 \Rightarrow 1 - \frac{27c^4}{16} = 0.$$

$$\boxed{c = \sqrt[4]{\frac{16}{27}} = \frac{2}{\sqrt[4]{27}}.}$$

Takeaways 3.8

- Intersections of conics can encode polynomial roots geometrically.
- Clearing denominators can convert a geometry problem into a quartic equation.
- Tangency corresponds to a repeated root, so the equations $P(x) = 0$ and $P'(x) = 0$ should be solved together.

Problem 3.9: A Cubic with a Double Root

The equation

$$x^3 - 5x^2 + mx - 3 = 0$$

has a double root. Find m.

Hint 3.9

Use $P(r) = 0$ and $P'(r) = 0$ for a double root r.

Solution 3.9

Let

$$P(x) = x^3 - 5x^2 + mx - 3.$$

If r is a double root, then

$$P(r) = 0, \qquad P'(r) = 0.$$

Since

$$P'(x) = 3x^2 - 10x + m,$$

we have

$$m = 10r - 3r^2.$$

Substitute into $P(r) = 0$:

$$r^3 - 5r^2 + r(10r - 3r^2) - 3 = 0.$$

So

$$2r^3 - 5r^2 + 3 = 0.$$

Factor:

$$(r - 1)(2r^2 - 3r - 3) = 0.$$

The easy integer root is $r = 1$, so

$$\boxed{m = 10(1) - 3(1)^2 = 7.}$$

Indeed, $x^3 - 5x^2 + 7x - 3 = (x - 1)^2(x - 3)$.

Takeaways 3.9

- Repeated-root parameter questions naturally produce a system in r and the parameter.
- A well-chosen repeated root often gives a simple integer value for the parameter.

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Problem 3.10: No Real Roots via Global Minimum

Prove that the polynomial below has no real roots:

$$P(x) = \frac{x^8}{8} + \frac{x^7}{7} + \frac{x^6}{6} + \frac{x^5}{5} + \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2} + x + \frac{11}{16}$$

Hint 3.10

- Differentiate $P(x)$ to find its stationary points. Do you recognize the geometric progression in $P'(x)$? Rewrite it using the sum formula.
- Find the exact x-coordinate of the only real turning point.
- Evaluate $P(x)$ at this turning point to find the global minimum. If the lowest possible point on an even-degree polynomial sits above the x-axis, what does that mean for its roots?

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Solution 3.10

1. Find the derivative to locate stationary points:

$$P'(x) = x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1$$

Recognizing this as a geometric series with common ratio x, we can rewrite it (for $x \neq 1$) as:

$$P'(x) = \frac{x^8 - 1}{x - 1}$$

2. Solve $P'(x) = 0$ for critical points:

$$\frac{x^8 - 1}{x - 1} = 0 \implies x^8 - 1 = 0 \quad (\text{where } x \neq 1)$$

The real solutions to $x^8 = 1$ are $x = 1$ and $x = -1$. Since $x \neq 1$ (which makes the denominator zero), our only valid stationary point is $x = -1$.

3. Determine the nature of the stationary point:

Analyze the sign of $P'(x)$ around $x = -1$:

- For $x < -1$ (e.g., $x = -2$): $P'(-2) = -85 < 0$ (Decreasing)
- For $-1 < x < 1$ (e.g., $x = 0$): $P'(0) = 1 > 0$ (Increasing)

Since the curve changes from decreasing to increasing, $x = -1$ is a local minimum. Because it is the *only* turning point and the polynomial has an even degree with a positive leading coefficient, it is the **global minimum**.

4. Evaluate the global minimum:

Substitute $x = -1$ into the original polynomial:

$$P(-1) = \frac{(-1)^8}{8} + \frac{(-1)^7}{7} + \cdots + \frac{(-1)^2}{2} + (-1) + \frac{11}{16}$$

$$P(-1) = \frac{1}{8} - \frac{1}{7} + \frac{1}{6} - \frac{1}{5} + \frac{1}{4} - \frac{1}{3} + \frac{1}{2} - 1 + \frac{11}{16}$$

Find a common denominator for the first eight fraction terms (840):

$$P(-1) = \left(\frac{105 - 120 + 140 - 168 + 210 - 280 + 420 - 840}{840}\right) + \frac{11}{16}$$

$$P(-1) = -\frac{533}{840} + \frac{11}{16}$$

Find a common denominator (1680) to add these final fractions:

$$P(-1) = -\frac{1066}{1680} + \frac{1155}{1680} = \mathbf{\frac{89}{1680}}$$

Conclusion:

Since $\frac{89}{1680} > 0$, the absolute lowest point of the curve is strictly above the x-axis. Therefore, $P(x) > 0$ for all real x, meaning the equation $P(x) = 0$ has **no real roots**.

Takeaways 3.10

- **Calculus as a Root-Finding Tool:** You do not need to solve an equation to prove it has no solutions. Proving the global minimum is strictly positive is a powerful Extension 1 technique.
- **The Geometric Polynomial:** The expression $x^n + x^{n-1} + \cdots + x + 1$ appears frequently. Immediately recognizing it as $\frac{x^{n+1}-1}{x-1}$ saves time and avoids messy factor-by-grouping methods.

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4 Conclusion

Polynomials reward systematic thinking as much as algebraic fluency. Many HSC Extension 1 questions become far more tractable once you identify the correct strategy: applying the factor theorem, writing out Vieta's formulas, or reading the graph multiplicity from first principles. As this booklet grows, it will serve as a compact reference for the most important polynomial patterns students encounter in Extension 1 examinations and beyond.

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A Appendices

A.1 Polynomial Remainder Theorem

Theorem A.1

If a polynomial $P(x)$ with real coefficients is divided by the linear factor $(x - a)$, the remainder is $P(a)$.

A.2 Factor Theorem

Theorem A.2

For a polynomial $P(x)$ with real coefficients, $x - a$ is a factor of $P(x)$ if and only if $P(a) = 0$. This is a special case of the polynomial remainder theorem.

A.3 Rational Root Theorem

Theorem A.3

The Rational Root Theorem states that if a polynomial

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

with integer coefficients has a rational root $x = \frac{p}{q}$ (where p and q are integers with a highest common factor of 1), then p is a factor of the constant term a_0 and q is a factor of the leading coefficient a_n.

The theorem is provided here as a good tip to find rational or integer roots of a polynomial with integer coefficients.

How it helps finding roots quickly

When solving polynomial equations $P(x) = 0$ or factoring polynomials, guessing a root can be time-consuming. The Rational Root Theorem narrows down the infinite number of possible rational roots to a finite list of candidates $\pm\frac{p}{q}$. By testing these candidates using the Factor Theorem, we can quickly locate a root and then use polynomial division to find the remaining factors.

Examples

Example A.1 (Simple Roots)

Consider the polynomial $P(x) = x^3 - 2x^2 - x + 2$. Here, the leading coefficient is $a_3 = 1$ and the constant term is $a_0 = 2$. The factors of $a_0 = 2$ are $p \in \{\pm1, \pm2\}$. The factors of $a_3 = 1$ are $q \in \{\pm1\}$. The possible rational roots $\frac{p}{q}$ are $\pm1, \pm2$. Testing $x = 1$, we get $P(1) = 1 - 2 - 1 + 2 = 0$. So $+1$ is a root. Testing $x = -1$, we get $P(-1) = -1 - 2 + 1 + 2 = 0$. So -1 is a root.

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Example A.2 (Fractional Roots)

Consider $P(x) = 2x^3 + 3x^2 - 8x + 3$. The leading coefficient is $a_3 = 2$ (factors $q \in \{\pm1, \pm2\}$) and the constant term is $a_0 = 3$ (factors $p \in \{\pm1, \pm3\}$). The possible rational roots are $\pm1, \pm3, \pm\frac{1}{2}, \pm\frac{3}{2}$. Testing $x = 1$, we get $P(1) = 2(1) + 3(1) - 8(1) + 3 = 0$. So 1 is a root. Testing $x = \frac{1}{2}$, we get $P\left(\frac{1}{2}\right) = 2\left(\frac{1}{8}\right) + 3\left(\frac{1}{4}\right) - 8\left(\frac{1}{2}\right) + 3 = \frac{1}{4} + \frac{3}{4} - 4 + 3 = 0$. So $\frac{1}{2}$ is a root.

A.4 Common Problem-Solving Transformations

Students often get stuck because they try to solve polynomials via brute force rather than recognising algebraic structures. This appendix details common substitution tricks:

1. Reciprocal Polynomials (Palindromic Coefficients)

Pattern: Polynomials like $ax^4 + bx^3 + cx^2 + bx + a = 0$.

Strategy: Divide by x^2 and use the substitution $u = x + \frac{1}{x}$. This immediately drops a degree-4 equation down to a solvable quadratic.

Example A.3

Solve $x^4 - 2x^3 - x^2 - 2x + 1 = 0$.

Solution: Divide by x^2 (since $x = 0$ is not a root):

$$x^2 - 2x - 1 - \frac{2}{x} + \frac{1}{x^2} = 0 \implies \left(x^2 + \frac{1}{x^2}\right) - 2\left(x + \frac{1}{x}\right) - 1 = 0.$$

Let $u = x + \frac{1}{x}$. Then $u^2 = x^2 + \frac{1}{x^2} + 2 \implies x^2 + \frac{1}{x^2} = u^2 - 2$. Substituting into the equation gives:

$$(u^2 - 2) - 2u - 1 = 0 \implies u^2 - 2u - 3 = 0 \implies (u - 3)(u + 1) = 0.$$

Thus $u = 3$ or $u = -1$. Solving $x + \frac{1}{x} = 3 \implies x^2 - 3x + 1 = 0$ gives $x = \frac{3\pm\sqrt{5}}{2}$. Solving $x + \frac{1}{x} = -1 \implies x^2 + x + 1 = 0$ gives no real solutions.

2. Biquadratic Polynomials (Even Powers)

Pattern: $P(x) = ax^4 + bx^2 + c$.

Strategy: Substitute $u = x^2$. Simple but highly common in HSC exams.

Example A.4

Find all real roots of $x^4 - 5x^2 + 4 = 0$.

Solution: Let $u = x^2$. The equation becomes $u^2 - 5u + 4 = 0$. Factoring gives $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Substituting back, $x^2 = 1 \implies x = \pm1$, and $x^2 = 4 \implies x = \pm2$.

3. Root Transformations (Shifting/Scaling)

Pattern: Given roots α, β, γ, find the equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ or $\alpha - k, \beta - k, \gamma - k$.

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Strategy: Instead of using Vieta's formulas to calculate the new coefficients, substitute $x = \frac{1}{u}$ or $x = u + k$ directly into the original polynomial.

Example A.5

Let $P(x) = x^3 - 4x + 1$ have roots α, β, γ. Find a polynomial with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$.
Solution: Substitute $x = \frac{1}{u}$ into $P(x) = 0$:

$$\left(\frac{1}{u}\right)^3 - 4\left(\frac{1}{u}\right) + 1 = 0.$$

Multiply by u^3 to clear the denominators:

$$1 - 4u^2 + u^3 = 0.$$

The required polynomial is $Q(x) = x^3 - 4x^2 + 1$.

A.5 Systems of Equations via Multiple Roots

A dedicated section on how to handle unknown coefficients (parameters like a, b, k) when given a multiple root.

The Pattern: "Find a and b given that $x^4 + ax^3 + bx - 4$ has a double root at $x = 2$."

The Strategy: Explicitly setting up the system $P(2) = 0$ and $P'(2) = 0$. This demonstrates how using calculus saves students from the nightmare of doing algebraic long division with parameters.

Example A.6

Find constants a and b given that $P(x) = x^4 + ax^3 + bx - 4$ has a double root at $x = 2$.
Solution: For a double root at $x = 2$, we require $P(2) = 0$ and $P'(2) = 0$. First, $P(2) = 2^4 + a(2^3) + b(2) - 4 = 16 + 8a + 2b - 4 = 12 + 8a + 2b$. Setting $P(2) = 0$ yields $4a + b = -6 \quad (1)$.
Next, find the derivative:

$$P'(x) = 4x^3 + 3ax^2 + b.$$

Evaluate at $x = 2$: $P'(2) = 4(2^3) + 3a(2^2) + b = 32 + 12a + b$. Setting $P'(2) = 0$ yields $12a + b = -32 \quad (2)$.
Subtract (1) from (2): $(12a - 4a) + (b - b) = -32 - (-6) \implies 8a = -26 \implies a = -\frac{13}{4}$.
Substitute $a = -\frac{13}{4}$ into (1): $4(-\frac{13}{4}) + b = -6 \implies -13 + b = -6 \implies b = 7$.
Therefore, $a = -\frac{13}{4}$ and $b = 7$.